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Hydraulic-pump flow-rate calibration system

Abstract

A hydraulic-pump flow-rate calibration system includes: a variable capacitance type hydraulic pump that supplies an operating fluid to a hydraulic actuator; a regulator that changes the dispense flow rate of the hydraulic pump according to a flow rate command signal; a flow rate detection device that detects the flow rate of the operating fluid; a control device that outputs the flow rate command signal to the regulator to control the regulator; and a calibration device that calculates an actual measurement characteristic of the dispense flow rate for the flow rate command signal, and performs, on a preset reference characteristic, calibration based on the actual measurement characteristic. The actual measurement characteristic is calculated as a result of the flow rate of the operating fluid being detected by the flow rate detection device during output of a predetermined flow rate command signal from the control device to the regulator.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2009/0077837	12/2008	Tozawa et al.	N/A	N/A
2012/0101744	12/2011	Matsuura	702/50	G01F 15/068
2021/0324609	12/2020	Takahashi	N/A	E02F 9/2285

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3379090	12/2017	EP	N/A
H11311203	12/1998	JP	N/A
2006129422	12/2005	WO	N/A
2017085996	12/2016	WO	N/A

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a hydraulic-pump flow-rate calibration system which, in the state where a hydraulic pump is connected to a hydraulic actuator, calibrates the dispense flow rate of the hydraulic pump.

BACKGROUND ART

(2) Construction equipment such as an excavator is capable of performing various tasks such as

digging by attachments such as a bucket provided on the construction equipment, and includes an actuator and a supply system in order to perform these tasks. Examples of the actuator include a hydraulic cylinder and a hydraulic motor. By being supplied with an operating fluid, for example, pressure oil, each of the hydraulic cylinder and the hydraulic motor operates in a direction corresponding to the flow direction of the pressure oil supplied thereto, at a speed corresponding to the flow rate of the pressure oil supplied thereto. Furthermore, a supply system is connected to the actuator, and the supply system includes a pump and a directional control valve. In the supply system, the pressure oil is dispensed from the pump in order to operate the actuator, and the directional control valve controls the flow direction and the flow rate of the pressure oil to be supplied from the pump to the actuator. This makes it possible to operate the actuator in a desired direction at a desired speed.

(3) In the supply system having such functions, a pump of the variable capacitance type is used, and the dispense flow rate of the pump is changed according to circumstances to improve the energy efficiency of the supply system. In order to meet such demands, for example, a swash plate pump is used as the pump of the variable capacitance type, and a regulator is configured as follows to rotate the swash plate of the swash plate pump at an angle. Specifically, the regulator rotates the swash plate at an angle corresponding to a signal pressure output from an electromagnetic proportional control valve, and the electromagnetic proportional control valve outputs the signal pressure corresponding to a signal (that is, an electric current) input thereto. In other words, the regulator can cause the pump to dispense the operating fluid at a flow rate corresponding to the signal input to the electromagnetic proportional control valve (that is, a flow rate corresponding to flow rate characteristics); in the supply system, the dispense flow rate of the pump can be electrically controlled.

(4) The supply system configured as described above varies from one product to another in terms of the flow rate characteristics of the regulator. Therefore, the flow rate characteristics are measured in pre-shipment inspection at manufacturing plants, etc., to inspect whether or not the flow rate characteristics are within a range of tolerance, and when the flow rate characteristics are not within the range of tolerance, a component of the regulator is replaced, for example, so that the flow rate characteristics fall in the range of tolerance. In this manner, accurate control of the dispense flow rate of the pump is enabled to further improve the energy efficiency of the supply system.

SUMMARY OF INVENTION

Technical Problem

(5) As mentioned above, at manufacturing plants, etc., the flow rate characteristics are measured in the pre-shipment inspection before shipment of the pump of the variable capacitance type, but in the inspection, the measurement is carried out under only one predetermined pressure condition. On the other hand, it is often the case that a pressure condition under which actual construction equipment or the like with the pump of the variable capacitance type mounted thereon is used does not necessarily match the pressure condition used in the pre-shipment inspection, meaning that the flow rate characteristics measured in the pre-shipment inspection are not reproduced with the actual equipment. In other words, there is an error between the flow rate characteristics measured in the pre-shipment inspection and the flow rate characteristics exhibited by actual equipment. Therefore, in order to eliminate such an error that occurs on actual equipment, it is desired that the dispense flow rate of the hydraulic pump be calibrated in the state where the hydraulic pump is mounted on actual equipment, to enable more precise control of the dispense flow rate.

(6) Thus, an object of the present invention is to provide a hydraulic-pump flow-rate calibration system capable of calibrating the dispense flow rate of a hydraulic pump in the state where the hydraulic pump is mounted on actual equipment.

Solution to Problem

(7) A hydraulic-pump flow-rate calibration system according to the present invention includes: a

hydraulic pump of a variable capacitance type that is connected to a hydraulic actuator and supplies an operating fluid to the hydraulic actuator, the hydraulic actuator operating at a speed corresponding to a flow rate of the operating fluid supplied to the hydraulic actuator; a regulator that changes a dispense flow rate of the hydraulic pump according to a flow rate command signal input to the regulator; a flow rate detection device that detects a flow rate of the operating fluid to be supplied to the hydraulic actuator; a control device that outputs the flow rate command signal to the regulator to control the regulator; and a calibration device that calculates an actual measurement characteristic of the dispense flow rate for the flow rate command signal, and performs, on a preset reference characteristic, calibration based on the actual measurement characteristic. The actual measurement characteristic is calculated as a result of the flow rate of the operating fluid to be supplied to the hydraulic actuator being detected by the flow rate detection device during output of a predetermined flow rate command signal from the control device to the regulator.

(8) According to the present invention, in the state where the hydraulic pump is connected to the hydraulic actuator, for example, in actual construction equipment or the like, the dispense flow rate of the hydraulic pump can be calibrated. This makes it possible to reduce variations in the operation of the hydraulic actuator from one machine to another when the operating fluid is supplied from the hydraulic pump to the hydraulic actuator.

(9) In the above invention, it is preferable that the hydraulic actuator be a hydraulic motor, and the flow rate detection device include a rotation sensor that detects a value corresponding to a rotational speed of an output shaft of the hydraulic motor and detect, on the basis of a result of the detection of the rotation sensor and a displacement of the hydraulic motor, the flow rate of the operating fluid to be supplied to the hydraulic motor.

(10) According to the above configuration, by estimating the detection flow rate using the rotation sensor, it is possible to calibrate the dispense flow rate of the hydraulic pump even when a flow rate sensor that detects a flow rate directly is not provided.

(11) In the above invention, it is preferable that the hydraulic motor cause a turning body rotatably provided on a structure to turn, the rotation sensor detect a speed of turning of the turning body as the value corresponding to the rotational speed of the output shaft of the hydraulic motor, and the flow rate detection device detect, on the basis of the speed of turning detected and the displacement of the hydraulic motor, the flow rate of the operating fluid to be supplied to the hydraulic motor.

(12) According to the above configuration, by detecting the speed of turning of the turning body, it is possible to calibrate the dispense flow rate of the hydraulic pump.

(13) In the above invention, it is preferable that the hydraulic-pump flow-rate calibration system further include a control unit that includes the calibration device and is provided on the turning body, and the rotation sensor be a gyroscope sensor and be embedded in the control unit.

(14) According to the above configuration, the gyroscope sensor embedded in the control unit can calculate the speed of turning of the turning body; thus, there is no need to additionally provide a rotation sensor, and thus an increase in the number of components can be minimized.

(15) A hydraulic-pump flow-rate calibration system according to the present invention includes: a first hydraulic pump of a variable capacitance type that is connected to a hydraulic actuator and supplies an operating fluid to the hydraulic actuator, the hydraulic actuator operating at a speed corresponding to a flow rate of the operating fluid supplied to the hydraulic actuator; a second hydraulic pump that is connected to the hydraulic actuator and supplies the operating fluid to the hydraulic actuator; a first regulator that changes a dispense flow rate of the first hydraulic pump according to a first flow rate command signal input to the first regulator; a switch valve that is connected to the first hydraulic pump, the second hydraulic pump, and the hydraulic actuator and connects one of the first hydraulic pump and the second hydraulic pump to the hydraulic actuator; a flow rate detection device that detects the flow rate of the operating fluid to be supplied to the hydraulic actuator; a control device that outputs the first flow rate command signal to the first regulator to control the first regulator; and a calibration device that calculates a first actual

measurement characteristic of the dispense flow rate of the first hydraulic pump for the first flow rate command signal, and performs, on a preset first reference characteristic, calibration based on the first actual measurement characteristic. The first actual measurement characteristic is calculated as a result of the first hydraulic pump and the hydraulic actuator being connected by the switch valve and the flow rate of the operating fluid to be supplied to the hydraulic actuator being detected by the flow rate detection device during output of a predetermined first flow rate command signal from the control device to the first regulator.

(16) According to the present configuration, in the state where two hydraulic pumps are connected to the hydraulic actuator, for example, in actual construction equipment or the like, the dispense flow rate of the first hydraulic pump can be calibrated. This makes it possible to reduce variations in the operation of the hydraulic actuator from one machine to another when the operating fluid is supplied from the first hydraulic pump to the hydraulic actuator.

(17) In the above invention, it is preferable that the hydraulic actuator be a hydraulic motor, and the flow rate detection device include a rotation sensor that detects a value corresponding to a rotational speed of an output shaft of the hydraulic motor and detect, on the basis of a result of the detection of the rotation sensor and a displacement of the hydraulic motor, the flow rate of the operating fluid to be supplied to the hydraulic motor.

(18) According to the above configuration, by estimating the detection flow rate using the rotation sensor, it is possible to calibrate the dispense flow rate of the hydraulic pump even when a flow rate sensor that detects a flow rate directly is not provided.

(19) In the above invention, it is preferable that the hydraulic motor cause a turning body rotatably provided on a structure to turn, the rotation sensor detect a speed of turning of the turning body as the value corresponding to the rotational speed of the output shaft of the hydraulic motor, and the flow rate detection device detect, on the basis of the speed of turning detected and the displacement of the hydraulic motor, the flow rate of the operating fluid to be supplied to the hydraulic motor.

(20) According to the above configuration, by detecting the speed of turning of the turning body, it is possible to calibrate the dispense flow rate of the hydraulic pump.

(21) In the above invention, it is preferable that the hydraulic-pump flow-rate calibration system further include a control unit that includes the calibration device and is provided on the turning body, and the rotation sensor be a gyroscope sensor and be embedded in the control unit.

(22) According to the above configuration, the gyroscope sensor embedded in the control unit can calculate the speed of turning of the turning body; thus, there is no need to additionally provide a rotation sensor, and thus an increase in the number of components can be minimized.

(23) In the above invention, it is preferable that the hydraulic-pump flow-rate calibration system further include a second regulator that changes, according to a second flow rate command signal input to the second regulator, a dispense flow rate of the second hydraulic pump that is of the variable capacitance type, the control device output the second flow rate command signal to the second regulator to control the second regulator, the calibration device calculate a second actual measurement characteristic of the dispense flow rate of the second hydraulic pump for the second flow rate command signal, and perform, on a preset second reference characteristic, calibration based on the second actual measurement characteristic, the second actual measurement characteristic be calculated as a result of the second hydraulic pump and the hydraulic actuator being connected by the switch valve and the flow rate of the operating fluid to be supplied to the hydraulic actuator being detected by the flow rate detection device during output of a predetermined second flow rate command signal to the second regulator.

(24) According to the present configuration, in the state where two hydraulic pumps are connected to the hydraulic actuator, for example, in actual construction equipment or the like, the dispense flow rates of both the first hydraulic pump and the second hydraulic pump can be calibrated. This makes it possible to reduce variations in the operation of the hydraulic actuator from one machine to another when the operating fluid is supplied from each hydraulic pump to the hydraulic actuator.

(25) In the above invention, it is preferable that the hydraulic-pump flow-rate calibration system further include: a replenishing unit connected to each of a supply passage formed between a first hydraulic actuator and the switch valve and a pump passage formed between the first hydraulic pump and the switch valve, the first hydraulic actuator being the hydraulic actuator; an exhaust valve connected to the pump passage and configured to be openable and closable, the exhaust valve being opened to discharge, to a tank, the operating fluid flowing in the pump passage; and an outflow rate detection device that detects a flow rate of the operating fluid flowing through the replenishing unit, and the switch valve be further connected to a second hydraulic actuator different from the first hydraulic actuator, and when the first hydraulic pump is connected to the first hydraulic actuator, the switch valve connect the second hydraulic pump to the second hydraulic actuator, and when the second hydraulic pump is connected to the first hydraulic actuator, the switch valve connect the first hydraulic pump to the second hydraulic actuator, and when the second hydraulic pump is connected to the first hydraulic actuator by the switch valve, the replenishing unit allow a flow directed from the supply passage to the pump passage to replenish the second hydraulic actuator with the operating fluid dispensed from the second hydraulic pump and blocks an opposite flow of the operating fluid, and the first actual measurement characteristic be calculated as a result of the first hydraulic pump and the first hydraulic actuator being connected by the switch valve and a flow rate of the operating fluid to be supplied to the first hydraulic actuator when the exhaust valve is closed being detected by the flow rate detection device during the output of the predetermined first flow rate command signal from the control device to the first regulator, and the second actual measurement characteristic be calculated on the basis of the flow rate detected by the flow rate detection device and an outflow rate detected by the outflow rate detection device, as a result of the second hydraulic pump and the first hydraulic actuator being connected by the switch valve and the flow rate of the operating fluid to be supplied to the first hydraulic actuator when the exhaust valve is open being detected by the flow rate detection device during the output of the predetermined second flow rate command signal to the second regulator.

(26) According to the above configuration, in a system including the replenishing unit, the dispense flow rate of the second hydraulic pump can be calibrated with high accuracy.

(27) In the above invention, it is preferable that the replenishing unit include a throttle, and the outflow rate detection device include a first pressure sensor that detects an outlet pressure of the first hydraulic pump and a second pressure sensor that detects an outlet pressure of the second hydraulic pump, and calculate the outflow rate on the basis of a difference between pressures detected by the first pressure sensor and the second pressure sensor.

(28) According to the above configuration, it is possible to obtain an accurate outflow rate when the operating fluid is supplied from the second hydraulic pump to the first hydraulic actuator, and thus the dispense flow rate of the second hydraulic pump can be calibrated with higher accuracy.

(29) In the above invention, it is preferable that the hydraulic-pump flow-rate calibration system further include: a second regulator that changes, according to a second flow rate command signal input to the second regulator, a dispense flow rate of the second hydraulic pump that is of the variable capacitance type; and a bypass passage connecting a supply passage formed between a first hydraulic actuator and the switch valve and a pump passage formed between the first hydraulic pump and the switch valve, the bypass passage including a bypass check valve that blocks a flow directed from the supply passage to the pump passage, the first hydraulic actuator being the hydraulic actuator, and the switch valve be further connected to a second hydraulic actuator different from the first hydraulic actuator, and when the first hydraulic pump is connected to the first hydraulic actuator, the switch valve connect the second hydraulic pump to the second hydraulic actuator, and when the second hydraulic pump is connected to the first hydraulic actuator, the switch valve connect the first hydraulic pump to the second hydraulic actuator, and the control device output the second flow rate command signal to the second regulator to control the second regulator, the calibration device calculate a second actual measurement characteristic of the

dispense flow rate of the second hydraulic pump for the second flow rate command signal, and perform, on a preset second reference characteristic, calibration based on the second actual measurement characteristic, the second actual measurement characteristic be calculated on the basis of a detection flow rate and a correction flow rate detected by the flow rate detection device, as a result of the first flow rate command signal serving as a reference being output to the first regulator, the second hydraulic pump being connected to the first hydraulic actuator by the switch valve, the operating fluid dispensed from the first hydraulic pump being supplied to the first hydraulic actuator via the bypass passage, the operating fluid dispensed from the second hydraulic pump being supplied to the first hydraulic actuator via the switch valve, and the flow rate of the operating fluid to be supplied to the first hydraulic actuator being detected by the flow rate detection device during output of a predetermined second flow rate command signal to the second regulator, and the correction flow rate be detected by the flow rate detection device when the first flow rate command signal serving as the reference is output from the control device to the first regulator and the first hydraulic pump is connected to the first hydraulic actuator by the switch valve.

(30) According to the above configuration, in the state where two hydraulic pumps are connected to the hydraulic actuator, for example, in actual construction equipment or the like, the dispense flow rates of both the first hydraulic pump and the second hydraulic pump can be calibrated. This makes it possible to reduce variations in the operation of the hydraulic actuator from one machine to another when the operating fluid is supplied from each hydraulic pump to the hydraulic actuator.

(31) In the above invention, it is preferable that the switch valve be capable of connecting both the first hydraulic pump and the second hydraulic pump to the hydraulic actuator, the calibration device calculate the second actual measurement characteristic of the dispense flow rate of the second hydraulic pump for the second flow rate command signal, and perform, on a preset second reference characteristic, calibration based on the second actual measurement characteristic, the second actual measurement characteristic be calculated on the basis of a detection flow rate and a correction flow rate detected by the flow rate detection device, as a result of the first flow rate command signal serving as a reference being output to the first regulator, both the first hydraulic pump and the second hydraulic pump being connected to the hydraulic actuator by the switch valve, and the flow rate of the operating fluid to be supplied to the hydraulic actuator being detected by the flow rate detection device during the output of the predetermined second flow rate command signal to the second regulator, and the correction flow rate be a flow rate of the operating fluid flowing through the hydraulic actuator when the first flow rate command signal serving as the reference is output from the control device to the first regulator and the first hydraulic pump is connected to the hydraulic actuator by the switch valve.

(32) According to the above configuration, in the state where two hydraulic pumps are connected to the hydraulic actuator, for example, in actual construction equipment or the like, the dispense flow rates of both the first hydraulic pump and the second hydraulic pump can be calibrated. This makes it possible to reduce variations in the operation of the hydraulic actuator from one machine to another when the operating fluid is supplied from each hydraulic pump to the hydraulic actuator.

(33) In the above invention, it is preferable that the calibration device correct, on the basis of an amount of leakage at the hydraulic actuator, the flow rate detected by the flow rate detection device, and calculate the actual measurement characteristic on the basis of the flow rate corrected.

(34) According to the above configuration, the dispense flow rate of each hydraulic pump can be calibrated with higher accuracy.

(35) In the above invention, it is preferable that the actual measurement characteristic be calculated on the basis of a plurality of flow rates detected by the flow rate detection device when a plurality of flow rate command signals different from each other are output.

(36) According to the above configuration, the dispense flow rate of each hydraulic pump can be calibrated with higher accuracy.

(37) In the above invention, it is preferable that when a predetermined condition is met, the calibration device calculate the actual measurement characteristic.

(38) According to the above configuration, when the condition is met, the hydraulic pump can be automatically calibrated, leading to an improvement in convenience.

Advantageous Effects of Invention

(39) With the present invention, the dispense flow rate of the hydraulic pump can be calibrated in the state where the hydraulic pump is mounted on actual equipment.

(40) The above object, other objects, features, and advantages of the present invention will be made clear by the following detailed explanation of preferred embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a perspective view illustrating an excavator on which a hydraulic drive system according to an embodiment of the present invention is mounted.

(2) FIG. 2 is a hydraulic circuit representing a hydraulic drive system according to Embodiment 1 which is mounted on the excavator illustrated in FIG. 1.

(3) FIG. 3 is a graph illustrating flow rate characteristics of a hydraulic pump in the hydraulic drive system illustrated in FIG. 2.

(4) FIG. 4 is a flowchart illustrating the flow of steps in a flow-rate calibration process which is performed by the hydraulic drive system illustrated in FIG. 2.

(5) FIG. 5 is a hydraulic circuit representing a hydraulic drive system according to each of Embodiments 2 to 4.

(6) FIG. 6 is a flowchart illustrating the flow of steps in a flow-rate calibration process which is performed by the hydraulic drive system illustrated in FIG. 5.

(7) FIG. 7 is a flowchart illustrating the flow of steps in a second pump calibration process which is performed by a hydraulic drive system according to Embodiment 2.

(8) FIG. 8 is a flowchart illustrating the flow of steps in a second pump calibration process which is performed by a hydraulic drive system according to Embodiment 3.

(9) FIG. 9 is a flowchart illustrating the flow of steps in a second pump calibration process which is performed by a hydraulic drive system according to Embodiment 4.

(10) FIG. 10 is a hydraulic circuit representing a hydraulic drive system according to Embodiment 5.

(11) FIG. 11 is a flowchart illustrating the flow of steps in a flow-rate calibration process which is performed by the hydraulic drive system illustrated in FIG. 10.

(12) FIG. 12 is a hydraulic circuit representing a hydraulic drive system according to another embodiment.

DESCRIPTION OF EMBODIMENTS

(13) Hereinafter, hydraulic drive systems **1**, **1A** to **1D** according to Embodiments 1 to 5, each of which is one example of the hydraulic-pump flow-rate calibration system according to the present invention, will be described with reference to the drawings. Note that the concept of directions mentioned in the following description is used for the sake of explanation; the orientations, etc., of elements according to the present invention are not limited to these directions. Each of the hydraulic drive systems **1**, **1A** to **1D** described below is merely one embodiment of the present invention. Thus, the present invention is not limited to the embodiments and may be subject to addition, deletion, and alteration within the scope of the essence of the present invention.

Embodiment 1

(14) Work equipment such as construction equipment is capable of performing various tasks using

an operating fluid (for example, oil). Examples of such work equipment include a crane, a wheel loader, and an excavator, and the following describes an example of application of an excavator 3 illustrated in FIG. 1. The excavator 3 is configured to be able to perform various tasks such as digging by an attachment, for example, a bucket 4, attached to a tip portion of the excavator 3. Furthermore, the excavator 3 includes a traveling device 5 such as a crawler in order to convey a dug material, and a turning body 6 is placed on the traveling device 5.

(15) A driver seat 6a for a driver to be seated thereon is formed on the turning body 6, and a boom 7 is provided on the turning body 6 so as to be able to swing vertically. An arm 8 is provided on a tip portion of the boom 7 so as to be able to swing vertically, and the bucket 4 is provided on a tip portion of the arm 8. In other words, the bucket 4 is provided on the turning body 6 via the boom 7 and the arm 8, and it is possible to raise and lower the bucket 4 by operating the boom 7 and the arm 8. Furthermore, the turning body 6 is configured to be able to turn with respect to the traveling device 5, which is the structure, and can cause the bucket 4 to move to any position in the 360-degree circle. The excavator 3 configured as just described includes a plurality of hydraulic actuators 11L, 11R, 12 to 15, for example, in order to move the traveling device 5, the turning body 6, the boom 7, the arm 8, and the bucket 4.

(16) Specifically, the excavator 3 includes a pair of left and right traveling hydraulic motors 11L, 11R, a turning hydraulic motor 12, a boom cylinder 13 (refer to FIG. 1), an arm cylinder 14 (refer to FIG. 1), and a bucket cylinder 15 (refer to FIG. 1). The pair of left and right traveling hydraulic motors 11L, 11R, which are so-called hydraulic motors, are supplied with the operating fluid, thereby drive a pair of left and right crawlers 5L, 5R, of the traveling device 5 to cause the excavator 3 to move forward and backward and change directions. The turning hydraulic motor 12 is provided on the turning body 6 in order to turn the turning body 6. The turning hydraulic motor 12, which is also a so-called hydraulic motor, is supplied with the operating fluid, thereby causing the turning body 6 to turn. The boom cylinder 13, the arm cylinder 14, and the bucket cylinder 15 are provided on the boom 7, the arm 8, and the bucket 4, respectively, and are supplied with the operating fluid and thereby extended and retracted, causing the boom 7, the arm 8, and the bucket 4 to swing, respectively. Thus, various hydraulic actuators 11L, 11R, 12 to 15 are configured to operate when supplied with the operating fluid, and in order to supply the operating fluid thereto, the excavator 3 includes the hydraulic drive system 1.

(17) [Hydraulic Drive System]

(18) As illustrated in FIG. 2, the hydraulic drive system 1 mainly includes two hydraulic pumps 21L, 21R, two regulators 23L, 23R, and a hydraulic supply device 24. The two hydraulic pumps 21L, 21R are, for example, tandem double pumps and can be driven by a shared input shaft 25. Note that the two hydraulic pumps 21L, 21R do not necessarily need to be the tandem double pumps and may be parallel double pumps or may each be a separately formed single pump. The number of hydraulic pumps included in the hydraulic drive system 1 is not necessarily limited to two and may be three or more. The two hydraulic pumps 21L, 21R configured as just described are connected to a drive source 26 such as an engine or an electric motor via the input shaft 25, and rotation of the input shaft 25 by the drive source 26 causes the operating fluid to be dispensed from the two hydraulic pumps 21L, 21R.

(19) The two hydraulic pumps 21L, 21R configured as described above are both variable-capacitance swash plate pumps and include swash plates 22L, 22R, respectively. Specifically, one of the two hydraulic pumps 21L, 21R, namely, the left hydraulic pump 21L, can change the dispense flow rate thereof by changing the tilt angle of the swash plate 22L, and the other of the two hydraulic pumps 21L, 21R, namely, the right hydraulic pump 21R, can change the dispense flow rate thereof by changing the tilt angle of the swash plate 22R. Furthermore, the regulators 23L, 23R are provided on the hydraulic pumps 21L, 21R, respectively, in order to change the tilt angles of the swash plates 22L, 22R of the hydraulic pumps 21L, 21R. The two regulators 23L, 23R can control the respective dispense flow rates of the hydraulic pumps 21L, 21R by adjusting

the tilt angles according to flow rate command signals input to these regulators.

(20) More specifically, each of the regulators **23L**, **23R** includes an electromagnetic proportional control valve (not illustrated in the drawings), and the electromagnetic proportional control valve outputs a signal pressure having a value corresponding to the input flow rate command signal. Accordingly, a servo piston (not illustrated in the drawings) of each of the regulators **23L**, **23R** moves to a position corresponding to the signal pressure. The aforementioned swash plates **22L**, **22R** are coupled to the servo pistons, and the swash plates **22L**, **22R** rotate according to movement of the servo pistons. Therefore, each of the swash plates **22L**, **22R** rotates through a tilt angle corresponding to the flow rate command signal; in other words, the operating fluid is dispensed from each of the hydraulic pumps **21L**, **21R** at a flow rate corresponding to the flow rate command signal. The operating fluid dispensed in this manner is supplied to the hydraulic actuators **11L**, **11R**, **12** to **15**, and in order to control the direction and the flow rate of the operating fluid that is supplied thereto, the hydraulic supply device **24** is connected to the two hydraulic pumps **21L**, **21R**.

(21) The hydraulic supply device **24** includes a plurality of directional control valves **31L**, **31R**, **32**. The directional control valves **31L**, **31R**, **32** are arranged corresponding to the aforementioned hydraulic actuators **11L**, **11R**, **12** to **15** and can control the flow and the flow rate of the operating fluid that is supplied to the corresponding hydraulic actuators **11L**, **11R**, **12** to **15**. More specifically, the hydraulic supply device **24** includes left and right traveling directional control valves **31L**, **31R** and a turning directional control valve **32** as the directional control valves corresponding to the hydraulic actuators **11L**, **11R**, **12**. The left and right traveling directional control valves **31L**, **31R** are arranged corresponding to the pair of left and right traveling hydraulic motors **11L**, **11R** and control the flow and the flow rate of the operating fluid that is supplied the corresponding traveling hydraulic motors **11L**, **11R**. On the other hand, the turning directional control valve **32** is arranged corresponding to the turning hydraulic motor **12** and controls the flow and the flow rate of the operating fluid that is supplied to the turning hydraulic motor **12**. Note that the hydraulic supply device **24** includes various directional control valves corresponding to the boom cylinder **13**, the arm cylinder **14**, the bucket cylinder **15**, and the like, in addition to the directional control valves **31L**, **31R**, **32**. For example, the directional control valve (not illustrated in the drawings) corresponding to the boom cylinder **13** is connected to a parallel passage **48** branching from a left pump passage **33L**. Thus, the hydraulic supply device **24** includes the plurality of directional control valves, but illustration and detailed description of the directional control valves other than the aforementioned three directional control valves **31L**, **31R**, **32** particularly related to the pump flow-rate calibration process to be described later will be omitted below.

(22) Furthermore, the hydraulic supply device **24** also includes a straight travel valve **30** to be described in detail later in addition to the aforementioned plurality of directional control valves **31L**, **31R**, **32**. Among the three directional control valves **31L**, **31R**, **32**, the two directional control valves **31L**, **32** except the right traveling directional control valve **31R** are connected to the straight travel valve **30**, which is one example of the switch valve. Furthermore, the straight travel valve **30** is connected to the left pump passage **33L** and the right pump passage **33R**, and is connected to the two hydraulic pumps **21L**, **21R** via the pump passages **33L**, **33R**. In other words, the two directional control valves **31L**, **32** are connected to the hydraulic pumps **21L**, **21R** via the straight travel valve **30**. Meanwhile, the right traveling directional control valve **31R** is connected to the right hydraulic pump **21R** so as to be parallel to the straight travel valve **30**. In other words, the right traveling directional control valve **31R** is connected to the right hydraulic pump **21R** without passing through the straight travel valve **30**; the right traveling directional control valve **31R** is configured as follows.

(23) The right traveling directional control valve **31R** is connected to the right pump passage **33R** and is also connected to the tank **27** and the right traveling hydraulic motor **11R** and can switch the connection thereof. More specifically, the right traveling directional control valve **31R** is what is

called a spool valve and includes a spool **31Ra**. The spool **31Ra** receives pilot pressures output from two different electromagnetic proportional control valves **31Rb**, **31Rc** provided at both ends of the spool **31Ra** and moves from a neutral position in either of predetermined opposite directions in accordance with the difference between the two pilot pressures received. Accordingly, the connection between the right traveling hydraulic motor **11R** and each of the right pump passage **33R** and the tank **27** is switched. Specifically, at the right traveling directional control valve **31R**, the right pump passage **33R** and the right traveling hydraulic motor **11R** are disconnected when the spool **31Ra** is in the neutral position. When the spool **31Ra** moves from the neutral position in either of the predetermined opposite directions, the right pump passage **33R** is connected to the right traveling hydraulic motor **11R**, and the operating fluid is supplied to the right traveling hydraulic motor **11R**. Furthermore, at the right traveling directional control valve **31R**, the flow direction of the operating fluid that is supplied to the right traveling hydraulic motor **11R** is switched according to the position of the spool **31Ra**, and by switching the flow direction, it is possible to change the direction of rotation of the right traveling hydraulic motor **11R**. Moreover, the degree of opening of the right traveling directional control valve **31R** is adjusted to be a degree of opening corresponding to the position of the spool **31Ra**, and the right traveling directional control valve **31R** controls the speed of the right traveling hydraulic motor **11R** by causing the operating fluid to flow to the right traveling hydraulic motor **11R** at a flow rate corresponding to the degree of opening.

(24) The right traveling directional control valve **31R** configured as described above is directly connected to the right hydraulic pump **21R** via the right pump passage **33R** as mentioned above. Meanwhile, the other directional control valves **31L**, **31R** are connected to the two hydraulic pumps **21L**, **21R** via the straight travel valve **30** as mentioned above, and the straight travel valve **30** is capable of switching between the hydraulic pumps **21L**, **21R** to be connected to the directional control valves **31L**, **31R**, according to the operating status of the excavator **3**. The straight travel valve **30** having such a function is configured as follows.

(25) The straight travel valve **30** is used to reduce the unevenness in the flow rates of the operating fluid flowing to the pair of left and right traveling hydraulic motors **11L**, **11R** at the time of operating an actuator and the like, for example, performing a boom operation, a turning operation, and the like while causing the excavator **3** to travel straight. In order to fulfill such a function, the straight travel valve **30** is capable of switching between the hydraulic pumps **21L**, **21R** to be connected to the two directional control valves **31L**, **32**, respectively. The straight travel valve **30** configured as just described is connected to the right pump passage **33R** so as to be parallel to the right traveling directional control valve **31R** as mentioned above, and is also connected to the left pump passage **33L**. Furthermore, a left supply passage **34L** and a right supply passage **34R** are connected to the straight travel valve **30**; the left traveling directional control valve **31L** is connected to the straight travel valve **30** via the left supply passage **34L**, and the turning directional control valve **32** is connected to the straight travel valve **30** via the right supply passage **34R**. The straight travel valve **30** disposed as just described switches the connection of each of these four passages **33L**, **33R**, **34L**, **34R** and switches between the hydraulic pumps **21L**, **21R** to be connected to the two directional control valves **31L**, **32**, respectively.

(26) More specifically, the straight travel valve **30** is what is called a spool valve and includes a spool **30a**. The spool **30a** can move along the axial line thereof; as a result of movement of the spool **30a**, the function of the straight travel valve **30** is switched. Specifically, the spool **30a** can move between a first position **A1** and a second position **A2**. When the spool **30a** is in the first position **A1**, the left pump passage **33L** is connected to the left supply passage **34L**, and the right pump passage **33R** is connected to the right supply passage **34R** (a first function). In contrast, when the spool **30a** is in the second position **A2**, the left pump passage **33L** is connected to the right supply passage **34R**, and the right pump passage **33R** is connected to the left supply passage **34L** (a second function). Furthermore, at the straight travel valve **30**, in the state where the spool **30a** is

located between the first position A1 and the second position A2, the connection of each of the four passages 33L, 33R, 34L, 34R changes as follows.

(27) Specifically, as the spool 30a moves from the first position A1 to the second position A2, the spool 30a increases the degree of opening between the left pump passage 33L and the right supply passage 34R. Furthermore, as the spool 30a moves from the first position A1 to the second position A2, the degree of opening between the right pump passage 33R and the left supply passage 34L increases. Moreover, at the straight travel valve 30, in the state where the spool 30a is located between the first position A1 and the second position A2, both the two pump passages 33L, 33R are connected to the two hydraulic pumps 21L, 21R (a merging function).

(28) In this manner, the straight travel valve 30 is designed to be able to switch the connection of each of the four passages 33L, 33R, 34L, 34R by changing the position of the spool 30a. Furthermore, a spring member 30b is provided on the spool 30a in order to change the position of the spool 30a. The spring member 30b is provided at one end of the spool 30a and biases the spool 30a in order to place the spool 30a in the first position A1. Furthermore, a switch command pressure acts on the other end of the spool 30a to withstand the force of the spring member 30b, and a switching electromagnetic proportional control valve 35 is connected to the straight travel valve 30 in order to exert the switch command pressure. The switching electromagnetic proportional control valve 35 outputs a switch command pressure having a value corresponding to a received switch command signal. The output switch command pressure is provided to the other end of the spool 30a as mentioned above, and the spool 30a is pressed with the pressing force corresponding to the switch command pressure.

(29) As described above, the basing force of the spring member 30b and the pressing force corresponding to the switch command pressure act on the ends of the spool 30a so as to oppose each other, and the spool 30a moves to a position where these forces are in balance. In other words, by adjusting the switch command pressure, it is possible to move the spool 30a between the first position A1 and the second position A2 and switch the connection destination of each of the two pump passages 33L, 33R to one of the supply passages 34L, 34R. The left traveling directional control valve 31L is connected to the left supply passage 34L, the connection destination of which is changeable as just described.

(30) The left traveling directional control valve 31L is connected to the left traveling hydraulic motor 11L and the tank 27 in addition to the left supply passage 34L and can switch the connection of each of the left traveling hydraulic motor 11L and the tank 27. More specifically, the left traveling directional control valve 31L is what is called a spool valve and includes a spool 31La. The spool 31La receives pilot pressures output from two different electromagnetic proportional control valves 31Lb, 31Lc provided at both ends of the spool 31La and moves from a neutral position in either of predetermined opposite directions in accordance with the difference between the two pilot pressures received. Accordingly, the connection between the left traveling hydraulic motor 11L and each of the left supply passage 34L and the tank 27 is switched. Specifically, at the left traveling directional control valve 31L, the left supply passage 34L and the left traveling hydraulic motor 11L are disconnected when the spool 31La is in the neutral position. When the spool 31La moves from the neutral position in either of the predetermined opposite directions, the left supply passage 34L is connected to the left traveling hydraulic motor 11L, and the operating fluid guided to the left supply passage 34L can be supplied to the left traveling hydraulic motor 11L. Furthermore, at the left traveling directional control valve 31L, the flow direction of the operating fluid that is supplied to the left traveling hydraulic motor 11L is switched according to the position of the spool 31La, and by switching the flow direction, it is possible to change the direction of rotation of the left traveling hydraulic motor 11L. Moreover, the degree of opening of the left traveling directional control valve 31L is adjusted according to the position of the spool 31La, and the left traveling directional control valve 31L controls the speed of the left traveling hydraulic motor 11L by causing the operating fluid to flow to the left traveling hydraulic motor 11L.

at a flow rate corresponding to the degree of opening. The left traveling directional control valve **31L** configured as just described is connected to the left supply passage **34L** as mentioned above. Meanwhile, the turning directional control valve **32** is connected to the right supply passage **34R**. (31) The turning directional control valve **32** is connected to the turning hydraulic motor **12** and the tank **27** in addition to the right supply passage **34R**. Note that a check valve **36** is provided between the right supply passage **34R** and the turning directional control valve **32**, and the flow of the operating fluid from the turning directional control valve **32** toward the right supply passage **34R** is blocked by the check valve **36**. The turning directional control valve **32** disposed as just described can switch the connection between the turning hydraulic motor **12** and each of the right supply passage **34R** and the tank **27**. More specifically, the turning directional control valve **32** is what is called a spool valve and includes a spool **32a**. The spool **32a** receives pilot pressures output from two different electromagnetic proportional control valves **32b**, **32c** provided at both ends of the spool **32a** and moves from a neutral position in either of predetermined opposite directions in accordance with the difference between the two pilot pressures received. Thus, the connection between the turning hydraulic motor **12** and each of the right supply passage **34R** and the tank **27** can be switched. Specifically, at the turning directional control valve **32**, the right supply passage **34R** and the turning hydraulic motor **12** are disconnected when the spool **32a** is in the neutral position. When the spool **32a** moves from the neutral position in either of the predetermined opposite directions, the right supply passage **34** is connected to the turning hydraulic motor **12**, and the operating fluid guided to the right supply passage **34** can be supplied to the turning hydraulic motor **12**. Furthermore, at the turning directional control valve **32**, the flow direction of the operating fluid that is supplied to the turning hydraulic motor **12** is switched according to the position of the spool **32a**, and by switching the flow direction, it is possible to change the direction of rotation of the turning hydraulic motor **12**. Moreover, the degree of opening of the turning directional control valve **32** is adjusted according to the position of the spool **32a**, and the turning directional control valve **32** controls the speed of the turning hydraulic motor **12** by causing the operating fluid to flow to the turning hydraulic motor **12** at a flow rate corresponding to the degree of opening.

(32) Note that the following elements are connected between the turning directional control valve **32** and the turning hydraulic motor **12**. Specifically, the turning directional control valve **32** is connected to the turning hydraulic motor **12** via two turning supply passages **37L**, **37R**, and relief valves **38L**, **38R** are connected to the two turning supply passages **37L**, **37R**, respectively. When the hydraulic pressure of the operating fluid flowing through the turning supply passages **37L**, **37R** connected to the two relief valves **38L**, **38R** exceeds a predetermined relief pressure, the two relief valves **38L**, **38R** discharge the operating fluid to the tank **27**. Furthermore, the two turning supply passages **37L**, **37R** are connected to the tank **27** via check valves **39L**, **39R** and are designed to be able to add the operating fluid from the tank **27** when there is a shortage of the operating fluid.

(33) Furthermore, the hydraulic supply device **24** includes: a bypass passage **40L** branching from the left supply passage **34L**; and a bypass passage **40R** branching from the right pump passage **33R**. In these two bypass passages **40L**, **40R**, the respective traveling directional control valves **31L**, **31R** are located. Specifically, the left traveling directional control valve **31L** is located in the left bypass passage **40L**, which is one of the bypass passages, and the degree of opening of the left bypass passage **40L** is adjusted according to the operation of the left traveling directional control valve **31L**. Meanwhile, the right traveling directional control valve **31R** is located in the right bypass passage **40R**, and the degree of opening of the right bypass passage **40R** is adjusted according to the operation of the right traveling directional control valve **31R**.

(34) Furthermore, in the hydraulic supply device **24**, a first replenishing passage **41** and a second replenishing passage **42** are formed in order to replenish each of the parallel passage **48** and the right supply passage **34R** with the operating fluid when the flow rate of the operating fluid in these passages is insufficient. The first replenishing passage **41** is formed to provide a bridge between the

left bypass passage **40L** and the parallel passage **48**, and the second replenishing passage **42** is formed to provide a bridge between the right bypass passage **40R** and the right supply passage **34R**. Furthermore, a check valve **43** is located in the first replenishing passage **41**. The check valve **43** guides the operating fluid from the left bypass passage **40L** to the parallel passage **48** and blocks the opposite flow of the operating fluid. In other words, the check valve **43** guides the operating fluid from the left bypass passage **40L** to the parallel passage **48** when the flow rate of the operating fluid in the parallel passage **48** is insufficient. Meanwhile, a check valve **44** is located in the second replenishing passage **42**. The check valve **44**, which is one example of the bypass check valve, guides the operating fluid from the right bypass passage **40R** to the right supply passage **34R** and blocks the opposite flow of the operating fluid. In other words, the check valve **44** guides the operating fluid from the right bypass passage **40R** to the right supply passage **34R** when the flow rate of the operating fluid in the right supply passage **34R** is insufficient. Furthermore, two unloader valves **45L**, **45R** are connected to the two pump passages **33L**, **33R**, respectively, and the two pump passages **33L**, **33R** are connected to the tank **27** via the corresponding unloader valves **45L**, **45R**.

(35) The two unloader valves **45L**, **45R** are, for example, spool valves, and include spools **45La**, **45Ra**. The two unloader valves **45L**, **45R** can adjust the degrees of openings of tank passages **46L**, **46R** connecting the corresponding pump passages **33L**, **33R** and the tank **27** by sliding the spools **45La**, **45Ra** and thereby control the flow rate of the operating fluid flowing to the supply passages **34L**, **34R** (that is, bleed-off control). Thus, the unloader valves **45L**, **45R** are designed to be able to adjust the degrees of openings of the tank passages **46L**, **46R** by sliding the spools **45La**, **45Ra**, in other words, changing the positions of the spools **45La**, **45Ra**; in order to change these positions, the unloader valves **45L**, **45R** include spring members **45Lb**, **45Rb**.

(36) The spring members **45Lb**, **45Rb** are provided at one end of the spools **45La**, **45Ra** and bias the spools **45La**, **45Ra** in order to close the tank passages **46L**, **46R**. Furthermore, left and right unloading command pressures act on the other end of the spools **45La**, **45Ra** to withstand the forces of the spring members **45Lb**, **45Rb**, and electromagnetic proportional control valves **45Lc**, **45Rc** are connected to the unloader valves **45L**, **45R** in order to output the left and right unloading command pressures. The electromagnetic proportional control valves **45Lc**, **45Rc** output the unloading command pressures having values corresponding to received unloading command signals. The output unloading command pressures are provided to the other end of the spools **45La**, **45Ra** as mentioned above, and the spools **45La**, **45Ra** are pressed with the pressing forces corresponding to the unloading command pressures.

(37) As described above, the basing forces of the spring members **45Lb**, **45Rb** and the pressing forces corresponding to the unloading command pressures act on the ends of the spools **45La**, **45Ra** so as to oppose each other, and the spools **45La**, **45R** move to positions where these forces are in balance. Therefore, by adjusting the unloading command pressures, it is possible to adjust the degrees of openings of the tank passages **46L**, **46R** and thus close the tank passages **46L**, **46R**.

(38) The hydraulic drive system **1** configured as described above further includes a control unit **50**, and the operation of the regulators **23L**, **23R**, the straight travel valve **30**, the directional control valves **31L**, **31R**, **32**, and the unloader valves **45L**, **45R** is controlled by the control unit **50**.

Furthermore, a turning operation device **51** and a traveling operation device **52** are electrically connected to the control unit **50**, which is the control device, and commands related to the operation of the hydraulic supply device **24** can be provided by these operation devices **51**, **52**. These operation devices **51**, **52** are provided on the excavator **3** (more specifically, the driver seat **6a**) in order to operate the turning hydraulic motor **12** and the pair of traveling hydraulic motors **11L**, **11R**; for example, the operation devices **51**, **52** include electric joysticks or remote control valves.

(39) More specifically, the turning operation device **51** includes a turning operation lever **51a** and is provided on the driver seat **6a** of the excavator **3** in order to operate the turning hydraulic motor **12**.

The turning operation lever **51a** can be pulled down; when the turning operation lever **51a** is pulled down, the turning operation device **51** outputs a signal to the control unit **50**. Meanwhile, the traveling operation device **52** is provided on the driver seat **6a** of the excavator **3** in order to operate the pair of left and right traveling hydraulic motors **11L**, **11R**. The traveling operation device **52** disposed as just described includes one pair of left and right foot pedals **52a**, **52b**; the foot pedal **52a** is provided corresponding to the left traveling hydraulic motor **11L**, and the foot pedal **52b** is provided corresponding to the right traveling hydraulic motor **11R**. Each of the foot pedals **52a**, **52b** can be operated, for example, by being stepped on with a foot; when the foot pedal **52a**, **52b** is operated, the traveling operation device **52** outputs a signal to the control unit **50**.

(40) The control unit **50** is designed to control the operation of the directional control valves **31L**, **31R**, **32** in accordance with the signals output from the operation devices **51**, **52**; the control unit **50** is configured as follows in order to control the operation of the directional control valves **31L**, **31R**, **32**. Specifically, the control unit **50** is electrically connected to the electromagnetic proportional control valves **31Lb**, **31Lc**, **31Rb**, **31Rc**, **32b**, **32c** provided on the directional control valves **31L**, **31R**, **32** and outputs command signals to the electromagnetic proportional control valves **31Lb**, **31Lc**, **31Rb**, **31Rc**, **32b**, **32c** in accordance with the signals output from the operation devices **51**, **52**. Furthermore, the control unit **50** is electrically connected to the switching electromagnetic proportional control valve **35** provided on the straight travel valve **30** as well and outputs a switch command signal to the switching electromagnetic proportional control valve **35**, for example, in accordance with the output signal of the traveling operation device **52**. Moreover, the control unit **50** is electrically connected to the electromagnetic proportional control valves **45Lc**, **45Rc**, which are connected to the unloader valves **45L**, **45R**, as well and outputs the unloading command signals to the electromagnetic proportional control valves **45Lc**, **45Rc** in accordance with the output signals of the operation devices **51**, **52**.

(41) Furthermore, the hydraulic drive system **1** includes the following elements. Specifically, the hydraulic drive system **1** includes a gyroscope sensor **60**. The gyroscope sensor **60**, which is the flow rate detection device, is a three-axis gyroscope sensor, for example, and is electrically connected to the control unit **50**. The gyroscope sensor **60** outputs, to the control unit **50**, signals corresponding to angular velocities about predetermined x-axis, y-axis, and z-axis, and the control unit **50** calculates the angular velocity about each axis on the basis of the signal from the gyroscope sensor **60**. The gyroscope sensor **60** configured as just described is provided in the turning body **6** so as to be housed in a casing **50a** of the control unit **50** such as that illustrated in FIG. **1**; in other words, the gyroscope sensor **60** is embedded in the control unit **50**. The gyroscope sensor **60** disposed as just described is designed to turn together with the turning body **6** at the time of turning of the turning body **60**, and the control unit **50** is capable of calculating the speed of turning of the turning body **6** on the basis of the signal output from the gyroscope sensor **60**.

(42) Furthermore, the hydraulic drive system **1** includes two pressure sensors **62L**, **62R**. One of the two pressure sensors **62L**, **62R**, that is, the left pressure sensor **62L**, is connected to the left pump passage **33L** and outputs a signal corresponding to the dispense pressure of the left hydraulic pump **21L** to the control unit **50**. The other pressure sensor, that is, the right pressure sensor **62R** is connected to the right pump passage **33R** and outputs a signal corresponding to the dispense pressure of the right hydraulic pump **21R** to the control unit **50**. Subsequently, the control unit **50** detects the dispense pressures of the two hydraulic pumps **21L**, **21R** on the basis of the signals output from the two pressure sensors **62L**, **62R**. In addition, the control unit **50** performs various calculations and stores a variety of information.

(43) [Operation of Hydraulic Drive System]

(44) In the hydraulic drive system **1** configured as described above, the control unit **50** controls the operation of the hydraulic supply device **24** in accordance with the operation performed on the operation devices **51**, **52** and operates the hydraulic actuators **11L**, **11R**, **12**. The operation of the control unit **50** performed to operate the hydraulic actuators **11L**, **11R**, **12** will be described below.

Specifically, when the turning operation lever **51a** is operated and a signal is output from the turning operation device **51**, the control unit **50** first operates the right unloader valve **45R** and closes the right tank passage **46R**. Furthermore, the control unit **50** outputs a turning command signal corresponding to the signal of the turning operation device **51** to the electromagnetic proportional control valve **32b** (or the electromagnetic proportional control valve **32c**) and operates the turning directional control valve **32**. At this time, the spool **30a** of the straight travel valve **30** is in the first position **A1**, and the turning directional control valve **32** is connected to the right hydraulic pump **21R** via the right pump passage **33R** and the right supply passage **34R**. Therefore, the operating fluid from the right hydraulic pump **21R** is supplied to the turning hydraulic motor **12**, and the turning hydraulic motor **12** rotates with the operating fluid. Furthermore, at the turning directional control valve **32**, the spool **32a** moves to a position corresponding to the amount of operation on the turning operation lever **51a**, and the turning directional control valve **32** opens with a degree of opening corresponding to the amount of the operation on the turning operation lever **51a**. Thus, the operating fluid is supplied to the turning hydraulic motor **12** at a flow rate corresponding to the degree of opening, allowing the turning body **6** to turn at a speed of turning that corresponds to the amount of the operation on the turning operation lever **51a**.

(45) Next, when only one of the pair of foot pedals **52a**, **52b**, for example, the left foot pedal **52a**, is operated and a signal is output from the traveling operation device **52**, the control unit **50** first operates the left unloader valve **45L** and closes the left tank passage **46L**. Furthermore, the control unit **50** outputs a traveling command signal corresponding to the signal of the traveling operation device **52** to the electromagnetic proportional control valve **31Lb** (or the electromagnetic proportional control valve **31Lc**) and operates the left traveling directional control valve **31L**. When only one of the pair of foot pedals **52a**, **52b** is operated, the spool **30a** of the straight travel valve **30** is in the first position **A1**, and the left traveling directional control valve **31L** is connected to the left hydraulic pump **21L** via the left pump passage **33L** and the left supply passage **34L**. Therefore, the operating fluid from the left hydraulic pump **21L** is supplied to the left traveling directional control valve **31L**, and the left traveling hydraulic motor **11L** operates with the operating fluid. Furthermore, at the left traveling directional control valve **31L**, the spool **31La** moves to a position corresponding to the amount of operation on the left foot pedal **52a**, and the left traveling directional control valve **31L** opens with a degree of opening corresponding to the amount of the operation on the left foot pedal **52a**. Thus, the operating fluid is supplied to the left traveling hydraulic motor **11L** at a flow rate corresponding to the degree of opening, allowing the left traveling hydraulic motor **11L** to rotate at a rotational speed that corresponds to the amount of the operation on the left foot pedal **52a**. In other words, it is possible to cause the left crawler **5L** to move at a speed corresponding to the amount of the operation on the left foot pedal **52a**.

(46) When only the right foot pedal **52b** is operated, the control unit **50** first operates the right unloader valve **45R** and closes the right tank passage **46R**. Furthermore, the control unit **50** outputs a traveling command signal to the electromagnetic proportional control valve **31Rb** (or the electromagnetic proportional control valve **31Rc**) and operates the right traveling directional control valve **31R**. Accordingly, the right traveling hydraulic motor **11R** rotates at a speed corresponding to the amount of operation on the right foot pedal **52b**, meaning that it is possible to cause the right crawler **5R** to move at a speed corresponding to the amount of the operation on the right foot pedal **52b**. In contrast, for example, in the case of causing the excavator **3** to travel straight while moving the boom, the turning body, and the like, that is, for example, in the case where both the foot pedals **52a**, **52b** are operated during the boom operation and the turning operation, the control unit **50** operates as follows.

(47) Specifically, when a signal is output from the traveling operation device **52** in the state where both the foot pedals **52a**, **52b** are operated, the control unit **50** outputs a switch command signal to the switching electromagnetic proportional control valve **35** connected to the straight travel valve **30** and causes the spool **30a** to move the second position **A2**. Thus, the function of the straight

travel valve **30** switches to the second function. This means that the left pump passage **33L** is connected to the right supply passage **34R**, and the right pump passage **33R** is connected to the left supply passage **34L**. Thus, both the left traveling directional control valve **31L** and the right traveling directional control valve **31R** are connected to the right hydraulic pump **21R**, and the turning directional control valve **32** is connected to the left hydraulic pump **21L**. Furthermore, the left traveling directional control valve **31L** and the right traveling directional control valve **31R** open with degrees of opening corresponding to the amounts of operation on the foot pedals **52a**, **52b**, and the operating fluid is guided to the hydraulic motors **11L**, **11R** at flow rates corresponding to the amounts of operation on the foot pedals **52a**, **52b**. Thus, it is possible to cause the hydraulic motors **11L**, **11R** to rotate at speeds corresponding to the amounts of operation on the foot pedals **52a**, **52b**, meaning that it is possible to cause the excavator **3** to travel straight at a speed corresponding to the amounts of operation on the foot pedals **52a**, **52b**.

(48) Connecting both the pair of the left and right traveling hydraulic motors **11L**, **11R** to one hydraulic pump **21R** at the time of traveling straight as mentioned above provides the following advantages. Specifically, in the case where the pair of the left and right traveling hydraulic motors **11L**, **11R** are connected to separate hydraulic pumps **21L**, **21R**, when the turning hydraulic motor **12** is operated together with the traveling hydraulic motors **11L**, **11R**, the operating fluid in the left hydraulic pump **21L** is guided to the turning hydraulic motor **12** as well. In this case, there will be a shortage of the operating fluid to be supplied to the left traveling hydraulic motor **11L**, and it is not possible to guide the operating fluid to the traveling hydraulic motor **11R** at a desired flow rate. Therefore, when both of the two foot pedals **52a**, **42b** are operated in order for straight travel, the flow rates of the operating fluid that is supplied to the traveling hydraulic motors **11L**, **11R** become uneven, causing a reduction in the straight-travel capability of the hydraulic excavator. In contrast, in the case where both the pair of the left and right traveling hydraulic motors **11L**, **11R** are connected to one hydraulic pump **21R**, the operating fluid from the right hydraulic pump **21R** is approximately evenly distributed to the traveling hydraulic motors **11L**, **11R** regardless of whether or not the turning hydraulic motor **12** is operated. Thus, the unevenness in the flow rates of the operating fluid that is supplied to the traveling hydraulic motors **11L**, **11R** can be reduced, and it is possible to improve the straight-travel capability of the excavator **3** at the time of traveling straight. Note that at the time of simultaneously operating the boom **7**, the arm **8**, and the bucket **4** except the turning body **6**, it is likewise possible to improve the straight-travel capability of the excavator **3**.

(49) In the hydraulic drive system **1** configured as described above, the control unit **50** controls the operation of the hydraulic supply device **24** in accordance with the operation performed on the operation devices **51**, **52** and operates the hydraulic actuators **11L**, **11R**, **12**. Furthermore, in order to operate the hydraulic actuators **11L**, **11R**, **12** at speeds corresponding to the amounts of operation on the operation devices **51**, **52** (for example, to operate the turning body **6** at a speed corresponding to the amount of the operation on the turning operation lever **51a**), the control unit **50** operates as follows. Specifically, the control unit **50** controls the degrees of opening of the directional control valves **31L**, **31R**, **32** and also controls the dispense flow rates of the hydraulic pumps **21L**, **21R** via the regulators **23L**, **23R**. More specifically, the hydraulic pumps **21L**, **21R** have flow rate characteristics such as those illustrated in FIG. 3. Here, the flow rate characteristics indicate the relationship between the dispense flow rate and the tilt angle (that is, the flow rate command signal); in FIG. 3, the horizontal axis represents the flow rate command signal (electric current), and the vertical axis represents the dispense flow rate. As illustrated in FIG. 3, the dispense flow rate of each of the hydraulic pumps **21L**, **21R** is a minimum flow rate Q_{min} when the flow rate command signal is less than or equal to I_{min} , and increases in proportion to the flow rate command signal when the flow rate command signal exceeds I_{min} . When the flow rate command signal is greater than or equal to I_{max} , the dispense flow rate of each of the hydraulic pumps **21L**, **21R** is a maximum flow rate Q_{max} .

(50) The control unit **50**, in which such flow rate characteristics (the solid line in FIG. **3**) are set in advance and stored, calculates, on the basis of the stored flow rate characteristics, namely, reference characteristics, flow rate command signals to be output to the regulators **23L**, **23R**, and causes the hydraulic pumps **21L**, **21R** to discharge the operating fluid at flow rates corresponding to the amounts of operation on the hydraulic pumps **21L**, **21R**. Meanwhile, the reference characteristics may be different from actual flow rate characteristics due to various causes. The control unit **50** including the calibration device functions to calibrate the stored reference characteristics in order to fill this gap. The following describes a hydraulic-pump flow-rate calibration process which is performed using the turning hydraulic motor **12**, which is one example of the first hydraulic actuator.

(51) [Hydraulic-Pump Flow-Rate Calibration Process]

(52) In the hydraulic drive system **1**, which is the hydraulic-pump flow-rate calibration system, first, the control unit **50** determines whether or not a predetermined calibration condition is met. The calibration condition is, for example, that a power switch for the excavator **3** shall be operated and thus electric power shall be supplied to the control unit **50** or that a calibration switch not illustrated in the drawings shall be operated and thus a calibration command shall be input to the control unit **50**. Alternatively, the calibration condition may be that a predetermined length of time shall have passed without the operation devices **51**, **52** being operated. When the calibration condition is met, the control unit **50** starts the flow rate calibration process such as that illustrated in FIG. **4**, and the processing transitions to Step **S1**.

(53) In Step **S1**, which is a first supply state switching step, the state of the hydraulic drive system **1** is switched to a first supply state in which the operating fluid dispensed from the right hydraulic pump **21R**, which is the first hydraulic pump, is supplied to the turning hydraulic motor **12**. Specifically, the control unit **50** outputs signals to the valves **30**, **31L**, **31R**, **32**, **45L**, **45R**, and controls the operation thereof in the following manner. More specifically, the control unit **50** closes the right tank passage **46R** by the right unloader valve **45R** to prevent bleeding off of the operating fluid that is dispensed from the right hydraulic pump **21R**. On the other hand, the left tank passage **46L** is completely open by the left unloader valve **45L**, and the entire amount of the operating fluid dispensed from the left hydraulic pump **21L** returns to the tank **27**. At the same time, the control unit **50** places the spool **30a** of the straight travel valve **30** in the first position **A1** to cause the operating fluid dispensed from the right hydraulic pump **21R** to be guided to the right supply passage **34R** via the straight travel valve **30**.

(54) Furthermore, the control unit **50** operates the turning directional control valve **32**, in other words, causes the spool **32a** of the turning directional control valve **32** to slide so that the operating fluid guided to the right supply passage **34R** is supplied to the turning hydraulic motor **12**. At this time, the spool **32a** is slid so that the degree of opening of the turning directional control valve **32** reaches the maximum degree. On the other hand, the control unit **50** places each of the spools **31La**, **31Ra** (including the spools of various directional control valves) of the directional control valves **31L**, **31R** (including various directional control valves corresponding to the boom cylinder **13**, the arm cylinder **14**, the bucket cylinder **15**, and the like) other than the turning directional control valve **32** in the neutral position, thereby preventing the operating fluid from flowing to the other hydraulic actuators such as the left traveling hydraulic motor **11L** (the second hydraulic actuator) and the right traveling hydraulic motor **11R**. In this manner, only the spool **32a** of the turning directional control valve **32** is slid to cause the entire operating fluid in the right hydraulic pump **21R** to be supplied to the turning hydraulic motor **12** alone. When the state of the hydraulic supply device **24** is switched to the first supply state in which the entire operating fluid in the right hydraulic pump **21R** is supplied to the turning hydraulic motor **12** alone in this manner, the processing transitions to Step **S2**.

(55) In Step **S2**, which is a command electric current setting step, a predetermined flow rate command signal **I1** (for example, the first flow rate command signal) which is set on the basis of

the flow rate characteristics stored in advance is output to the right regulator **23R** (for example, the first regulator) provided on the right hydraulic pump **21R** (for example, the first hydraulic pump). Here, the flow rate command signal **I1** is set in advance to satisfy $I_{min} \leq I1 \leq I_{max}$, on the basis of the aforementioned reference characteristics of the right hydraulic pump **21R**, namely, the first reference characteristics (refer to the solid line in FIG. 3), and the set flow rate command signal **I1** is output to the right regulator **23R**. Accordingly, the swash plate **22R** of the right hydraulic pump **21R** rotates through a tilt angle corresponding to the flow rate command signal **I1**, and the operating fluid is dispensed from the right hydraulic pump **21R** at a flow rate corresponding to the flow rate command signal **I1**. Subsequently, when the entire amount of the operating fluid is supplied to the turning hydraulic motor **12** via the straight travel valve **30** and the turning directional control valve **32**, the processing transitions to Step **S3**.

(56) In Step **S3**, which is a turning speed detection step, the speed of turning of the turning body **6** is detected. Specifically, the control unit **50** detects the speed of turning of the turning body **6** on the basis of the signal output from the gyroscope sensor **60**. Note that in the present embodiment, the gyroscope sensor **60** is mounted on the turning body **6** so that the z-axis of the gyroscope sensor **60** is substantially parallel to the pivot of the turning body **6**, and the control unit **50** calculates the speed of turning of the turning body **60** by detecting an angular velocity about the z-axis. However, the method for calculating the speed of turning of the turning body **6** is not limited to the aforementioned method; the speed of turning may be calculated on the basis of angular velocities about two or three axes detected on the basis of the signals output from the gyroscope sensor **60**. When the speed of turning of the turning body **6** is detected in this manner, the processing transitions to Step **S4**.

(57) In Step **4**, which is a turning flow rate calculation step, the flow rate of the operating fluid supplied to the turning hydraulic motor **12** at the time of turning, namely, a turning flow rate, is calculated. Specifically, the control unit **50** stores, in advance, a swept volume (displacement) of the turning hydraulic motor **12** and a speed reduction ratio between the turning hydraulic motor **12** and the turning body **6**, and calculates the turning flow rate on the basis of said swept volume and the speed of turning calculated in Step **S3**. More specifically, the turning flow rate is calculated by multiplying the speed of turning calculated in Step **S3** by the swept volume. When the turning flow rate is calculated, the processing transitions to Step **S5**.

(58) In Step **S5**, which is a first calibration point obtainment step, an actual dispense flow rate of the right hydraulic pump **21R** is calculated, and a calibration point for the right hydraulic pump **21R** is obtained on the basis of the calculated actual dispense flow rate. Specifically, the control unit **50** calculates the dispense flow rate of the right hydraulic pump **21R** on the basis of the turning flow rate calculated in Step **S4**, but, first, for this purpose, calculates an amount of leakage of the operating fluid at the turning hydraulic motor **12**, namely, a motor leakage amount. The motor leakage amount changes according to the dispense pressure of the operating fluid supplied to the turning hydraulic motor **12**, and the control unit **50** calculates the motor leakage amount on the basis of the dispense pressure of the right hydraulic pump **21R** and motor efficiency characteristics of the turning hydraulic motor **12**. Here, the dispense pressure of the right hydraulic pump **21R** is detected on the basis of the signal from the right pressure sensor **62R**, and the motor efficiency characteristics of the turning hydraulic motor **12** (characteristics which are related to the usage ratio of the supplied flow and change according to the pressure) is stored in the control unit **50** in advance. When the control unit **50** calculates the motor leakage amount, the control unit **50** adds the calculated motor leakage amount to the turning flow rate. Thus, the dispense flow rate (=the turning flow rate+the motor leakage amount) is calculated.

(59) Note that the motor leakage amount does not necessarily need to be calculated on the basis of the dispense pressure of the right hydraulic pump **21R** and may be set to a constant value on the basis of the motor efficiency characteristics of the turning hydraulic motor **12**. Furthermore, in calculating the dispense flow rate, the motor leakage amount does not necessarily need to be

referred to, and the dispense flow rate may be set equal to the turning flow rate. These two cases (specifically, the case where neither the pressure nor the motor leakage amount is referred to) are preferred when there is no need to calibrate the flow rate characteristics on the basis of a more accurate dispense flow rate; in the case where it is preferable to calibrate the first reference characteristics on the basis of a more accurate dispense flow rate, it is preferable that the motor leakage amount be calculated on the basis of the dispense pressure and the motor efficiency characteristics as mentioned above. The same holds true for the dispense flow rate of the left hydraulic pump **21L** to be described later.

(60) When the control unit **50** calculates the dispense flow rate, the control unit **50** stores the calculated dispense flow rate in association with the flow rate command signal **I1** set in Step **S2**. For example, in the case where the dispense flow rate applied in response to the flow rate command signal **I1** is high compared to the first reference characteristics (the solid line in FIG. **3**), a calibration point **71** is obtained, as illustrated in FIG. **3**. When the first calibration point, namely, the calibration point **71**, is calculated in this manner, the processing transitions to Step **S6**.

(61) In Step **S6**, which is a number-of-calibration-points checking step, whether or not two or more calibration points have been obtained is determined at the time of calibrating the first reference characteristics. Note that the number of calibration points to be obtained may be three or more. When the number of calibration points obtained is determined as one, the processing returns to Step **S2**, and the dispense flow rate of the right hydraulic pump **21R** to be applied in response to a flow rate command signal **I2** (the first flow rate command signal) having a value different from the value of the flow rate command signal **I1** is calculated. Specifically, the control unit **50** outputs, to the right regulator **23R**, a flow rate command signal **I2** ($I_{min} \leq I2 \leq I_{max}$) having a value different from the value of the flow rate command signal **I1** in Step **S2**. When the control unit **50** outputs the set flow rate command signal **I2** to the right regulator **23R**, then the control unit **50** detects the speed of turning (Step **S3**) and further calculates the turning flow rate on the basis of the speed of turning detected in Step **S3** (Step **S4**). Furthermore, the control unit **50** calculates the dispense flow rate on the basis of the turning flow rate detected in Step **S4**, and stores the calculated dispense flow rate and the flow rate command signal **I2** in association with each other. When the second calibration point, namely, a calibration point **72**, is obtained in this manner (refer to FIG. **3**), the processing transitions to Step **S7**.

(62) In Step **S7**, which is a first pump flow rate calibration step, the first reference characteristics are calibrated on the basis of the two calibration points **71**, **72** obtained in Step **S5**. Specifically, in the range where the flow rate Q satisfies the relationship: $Q_{min} \leq Q \leq Q_{max}$, a straight line passing through the two calibration points **71**, **72** (refer to the dot-dashed line in FIG. **3**) is calculated as first actual measurement characteristics, which are actual flow rate characteristics of the right hydraulic pump **21R**. More specifically, the control unit **50** calculates, on the basis of the two calibration points **71**, **72**, a slope and an intercept of the first actual measurement characteristics in the range $Q_{min} \leq Q \leq Q_{max}$, calculates the first actual measurement characteristics, and sets the calculated first actual measurement characteristics as new first reference characteristics. When the first reference characteristics are calibrated on the basis of the first actual measurement characteristics in this manner, the processing transitions to Step **S8**.

(63) In Step **S8**, which is a second supply state switching step, the state of the hydraulic drive system **1** is switched to a second supply state in which the operating fluid dispensed from the left hydraulic pump **21L**, which is the second hydraulic pump, is supplied to the turning hydraulic motor **12**. Specifically, the control unit **50** outputs signals to the valves **30**, **31L**, **31R**, **32**, **45L**, **45R**, and controls the operation thereof in the following manner. More specifically, the control unit **50** closes the left tank passage **46L** by the left unloader valve **45L** to prevent bleeding off of the operating fluid that is dispensed from the left hydraulic pump **21L**. On the other hand, the right tank passage **46R** is completely open by the right unloader valve **45R**, and the entire amount of the operating fluid dispensed from the right hydraulic pump **21R** returns to the tank **27**. At the same

time, the control unit **50** places the spool **30a** of the straight travel valve **30** in the second position **A2** to cause the operating fluid dispensed from the left hydraulic pump **21L** to be guided to the right supply passage **34R** via the straight travel valve **30**. Furthermore, as in Step **S2**, the control unit **50** slides only the spool **32a** of the turning directional control valve **32** to cause the entire operating fluid in the left hydraulic pump **21L** to be supplied to the turning hydraulic motor **12** alone. On the other hand, the control unit **50** places each of the spools **31La**, **31Ra** (including the spools of various directional control valves) of the directional control valves **31L**, **31R** (including various directional control valves corresponding to the boom cylinder **13**, the arm cylinder **14**, the bucket cylinder **15**, and the like) other than the turning directional control valve **32** in the neutral position, thereby preventing the operating fluid from flowing to the other hydraulic actuators such as the left traveling hydraulic motor **11L** (the second hydraulic actuator) and the right traveling hydraulic motor **11R**. When the state of the hydraulic supply device **24** is switched to the second supply state in which the entire operating fluid in the left hydraulic pump **21L** is supplied to the turning hydraulic motor **12** alone in this manner, the processing transitions to Step **S9**.

(64) In Step **S9**, which is the command electric current setting step, a predetermined flow rate command signal **I3** (for example, the second flow rate command signal) which is set on the basis of the flow rate characteristics stored in advance is output to the left regulator **23L** (for example, the second regulator) provided on the left hydraulic pump **21L** (for example, the second hydraulic pump). Here, as with the flow rate command signal **I1** mentioned above, the flow rate command signal **I3** is set in advance to satisfy $I_{min} \leq I3 \leq I_{max}$, on the basis of the reference characteristics for the left hydraulic pump **21L**, namely, the second reference characteristics (refer to the solid line in FIG. 3), and the set flow rate command signal **I3** is output to the left regulator **23L**. Note that in the present embodiment, the same reference characteristics are set for the two hydraulic pumps **21L**, **21R**, but the reference characteristics do not necessarily need to be the same, and different reference characteristics may be set in advance. Furthermore, in the present embodiment, the flow rate command signal **I3** is set to a value different from the value of the flow rate command signal **I1**, but may be set to the same value as the value of the flow rate command signal **I1**. As a result of the flow rate command signal **I3** being output to the left regulator **23L**, the swash plate **22L** of the left hydraulic pump **21L** rotates through a tilt angle corresponding to the flow rate command signal **I3**, and the operating fluid is dispensed from the left hydraulic pump **21L** at a flow rate corresponding to the flow rate command signal **I3**. Subsequently, when the entire amount of the operating fluid is supplied to the turning hydraulic motor **12** via the straight travel valve **30** and the turning directional control valve **32**, the processing transitions to Step **S10**.

(65) In Step **S10**, which is the turning speed detection step, the speed of turning of the turning body **6** is detected as in Step **S3**. Specifically, the control unit **50** detects the speed of turning of the turning body **6** on the basis of the signal output from the gyroscope sensor **60**, and when the speed of turning of the turning body **6** is calculated, the processing transitions to Step **S11**. Furthermore, in Step **S11**, which is the turning flow rate calculation step, the turning flow rate of the turning hydraulic motor **12** at the time of turning is calculated as in Step **S4**. Specifically, the control unit **50** calculates the turning flow rate on the basis of the swept volume (displacement) of the turning hydraulic motor **12** and the speed reduction ratio between the turning hydraulic motor **12** and the turning body **6**, which are stored in advance, and the speed of turning calculated in Step **S10**, and when the turning flow rate is calculated, the processing transitions to Step **S12**.

(66) In Step **S12**, which is a second calibration point obtainment step, an actual dispense flow rate of the left hydraulic pump **21L** is calculated, and a calibration point for the left hydraulic pump **21L** is obtained on the basis of the calculated actual dispense flow rate. Specifically, the control unit **50** calculates the dispense flow rate of the left hydraulic pump **21L** on the basis of the turning flow rate calculated in Step **S11**, but, first, for this purpose, detects the dispense pressure of the left hydraulic pump **21L** on the basis of the signal from the left pressure sensor **62L**. Subsequently, the control unit **50** calculates the motor leakage amount of the turning hydraulic motor **12** on the basis

of the detected dispense pressure of the left hydraulic pump **21L** and the motor efficiency characteristics of the turning hydraulic motor **12**. Lastly, the control unit **50** calculates a dispense flow rate by adding the calculated motor leakage amount to the turning flow rate. When the dispense flow rate is calculated, the control unit **50** stores the calculated dispense flow rate in association with the flow rate command signal **I3** set in Step **S9**. For example, in the case where the dispense flow rate applied in response to the flow rate command signal **I3** is low compared to the second reference characteristics (the solid line in FIG. 3), a calibration point **73** is obtained, as illustrated in FIG. 3. When the first calibration point, namely, a calibration point **73**, is obtained in this manner, the processing transitions to Step **S13**.

(67) In Step **S13**, which is the number-of-calibration-points checking step, whether or not two or more calibration points have been obtained is determined at the time of calibrating the second reference characteristics. Note that the number of calibration points to be obtained may be three or more. When the number of calibration points obtained is determined as one, the processing returns to Step **S9**, and the dispense flow rate of the left hydraulic pump **21L** to be applied in response to a flow rate command signal **I4** (the second flow rate command signal) having a value different from the value of the flow rate command signal **I3** is calculated. Specifically, the control unit **50** outputs, to the left regulator **23L**, a flow rate command signal **I4** ($I_{min} \leq I4 \leq I_{max}$) having a value different from the value of the flow rate command signal **I3** in Step **S9**. Note that in the present embodiment, the flow rate command signal **I4** is set to a value different from the value of the flow rate command signal **I2**, but may be set to the same value as the value of the flow rate command signal **I1**. When the control unit **50** outputs the set flow rate command signal **I4** to the left regulator **23L**, then the control unit **50** detects the speed of turning (Step **S10**) and further calculates the turning flow rate on the basis of the speed of turning detected in Step **S10** (Step **S11**). Furthermore, the control unit **50** calculates the dispense flow rate on the basis of the turning flow rate detected in Step **S11**, and stores the calculated dispense flow rate and the flow rate command signal **I4** in association with each other. When the second calibration point, namely, a calibration point **74**, is obtained in this manner (refer to FIG. 3), the processing transitions from Step **S13** to Step **S14**.

(68) In Step **S14**, which is a second pump flow rate calibration step, the second reference characteristics are calibrated on the basis of the two calibration points **73**, **74** obtained in Step **S12**. Specifically, in the range where the flow rate Q satisfies the relationship: $Q_{min} \leq Q \leq Q_{max}$, a straight line passing through the two calibration points **73**, **74** (refer to the double-dot-dashed line in FIG. 3) is calculated as second actual measurement characteristics, which are actual flow rate characteristics of the left hydraulic pump **21L**. More specifically, the control unit **50** calculates, on the basis of the two calibration points **73**, **74**, a slope and an intercept of the second actual measurement characteristics in the range $Q_{min} \leq Q \leq Q_{max}$, calculates the second actual measurement characteristics, and sets the calculated second actual measurement characteristics as new second reference characteristics. When the second reference characteristics are calibrated on the basis of the second actual measurement characteristics in this manner, the flow rate calibration process ends.

(69) Thus, the hydraulic drive system **1** performs the flow rate calibration process described above and is capable of calibrating the flow rate characteristics of the two hydraulic pumps **21L**, **21R** in the state where the hydraulic drive system **1** is mounted on the excavator **3**. Therefore, in the excavator **3** with the hydraulic drive system **1** mounted thereon, the dispense flow rates of the two hydraulic pumps **21L**, **21R** can be controlled with high accuracy. Furthermore, the hydraulic drive system **1** can calculate the dispense flow rates of the two hydraulic pumps **21L**, **21R** on the basis of the speed of turning detected by the gyroscope sensor **60** and calibrate the flow rate characteristics on the basis of the calculated dispense flow rates. This means that in the hydraulic drive system **1**, the flow rate characteristics of the two hydraulic pumps **21L**, **21R** can be calibrated without addition of a flow rate sensor, and it is possible to minimize an increase in the number of components for the purpose of calibration.

Embodiment 2

(70) A hydraulic drive system **1A** according to Embodiment 2 is similar in configuration to the hydraulic drive system **1** according to Embodiment 1, as illustrated in FIG. 5. Therefore, the configuration of the hydraulic drive system **1A** according to Embodiment 2 will be described focusing on differences from the hydraulic drive system **1** according to Embodiment 1; elements that are the same as those of the hydraulic drive system **1** according to Embodiment 1 share the same reference signs, and as such, description of the elements will be omitted.

(71) A hydraulic supply device **24A** in the hydraulic drive system **1A** according to Embodiment 2 further includes a replenishing unit **47** in addition to the configuration of the hydraulic supply device **24** in the hydraulic drive system **1** according to Embodiment 1, and the replenishing unit **47** has the following function. Specifically, when the flow rate of the operating fluid flowing to the right pump passage **33R** is insufficient, the replenishing unit **47** guides the operating fluid from the right supply passage **34R** to the right pump passage **33R** to replenish the right pump passage **33R** with the operating fluid. More specifically, the replenishing unit **47** include a replenishing passage **47a**, a throttle **47b**, and a check valve **47c**. The replenishing passage **47a** is formed to provide a bridge between the right bypass passage **34R** and the right pump passage **33R**. Furthermore, in the replenishing passage **47a**, the throttle **47b** and the check valve **47c** are located; the throttle **47b** and the check valve **47c** are arranged in the replenishing passage **47a** in the stated order from the right supply passage **34R** side. The check valve **47c** disposed as just described allows the flow of the operating fluid from the right supply passage **34R** toward the right pump passage **33R** and blocks the opposite flow of the operating fluid.

(72) The hydraulic drive system **1A** configured as described above operates in substantially the same manner as the hydraulic drive system **1** according to Embodiment 1, but is different from the hydraulic drive system **1** according to Embodiment 1 as follows. Specifically, for example, when both the foot pedals **52a**, **52b** are operated during the boom operation and the turning operation, both the two hydraulic motors **11L**, **11R** are connected to the right hydraulic pump **21R**. This means that the operating fluid is supplied from the right hydraulic pump **21R** to the two hydraulic motors **11L**, **11R**. Therefore, when the amounts of operation on the foot pedals **52a**, **52b** are both great, the dispense flow rate of the right hydraulic pump **21R** alone may be insufficient at the time of supplying the operating fluid to the two hydraulic motors **11L**, **11R**. In such a case, the hydraulic drive system **1A** is capable of supplementing the insufficient flow rate by supplying the operating fluid from the right supply passage **34R** to the right pump passage **33R** via the replenishing unit **47**.

(73) The hydraulic drive system **1A** having such a function can further calibrate the flow rate characteristics of the two hydraulic pumps **21L**, **21R** in substantially the same flow rate calibration process as with the hydraulic drive system **1** according to Embodiment 1. However, since the replenishing unit **47** is provided, a portion of the operating fluid dispensed from the left hydraulic pump **21L** returns from the replenishing unit **47** to the tank **27** at the time of supplying the operating fluid from the left hydraulic pump **21L** to the turning hydraulic motor **12** in Steps **S9** to **S11**, meaning that it is not possible to accurately calculate the dispense flow rate of the left hydraulic pump **21L**. Thus, in order to accurately calculate the dispense flow rate of the left hydraulic pump **21L** and obtain more accurate flow rate characteristics of the two hydraulic pumps **21L**, **21R**, a control unit **50A** in the hydraulic drive system **1A** performs the following flow rate calibration process. Specifically, the control unit **50A** determines whether or not a predetermined calibration condition is met, and when the calibration condition is met, performs a flow rate calibration process such as that illustrated in FIG. 6. When the flow rate calibration process is performed, the processing transitions to Step **S1**, then the control unit **50A** performs Steps **S1** to **S5** to calibrate the flow rate of the right hydraulic pump **21R**, which is the first hydraulic pump, as with the hydraulic system **1** according to Embodiment 1.

(74) Specifically, when the flow rate calibration process is started, first, the state of the hydraulic drive system **1A** is switched to the first supply state (Step **S1**), and then the flow rate command

signal **I1** is set and output to the right regulator **23R** (Step **S2**). After the output, the speed of turning is detected (Step **S3**), and the turning flow rate is calculated on the basis of the speed of turning detected in Step **S3** (Step **S4**). Furthermore, the control unit **50A** calculates the dispense flow rate on the basis of the turning flow rate detected in Step **S4** and stores the calculated flow rate and the flow rate command signal **I1** in association with each other, in other words, obtains the calibration point **71** (refer to FIG. **3**) (Step **S5**). Moreover, since the obtained calibration point is the first calibration point, the processing returns from Step **S6** to Step **S2**, the flow rate command signal **12** is output to the right regulator **23R**, and the second calibration point, namely, the calibration point **72**, is obtained (Steps **S3** to **S5**). Subsequently, when it is determined that the two calibration points **71**, **72** have been obtained (Step **S6**), the first actual measurement characteristics are calculated on the basis of the two calibration points **71**, **72**, and the calculated first actual measurement characteristics are set as new first reference characteristics (Step **S7**). When the first reference characteristics are calibrated on the basis of the first actual measurement characteristics in this manner, the processing transitions to Step **S20**. In Step **S20**, the second pump calibration process such as that illustrated in FIG. **7** is performed, and the processing transitions to Step **S21**.

(75) In Step **S21**, which is a minimum tilt angle switching step, the swash plate **22R** of the right hydraulic pump **21R** rotates up to the minimum tilt angle. Specifically, the control unit **50A** sets a flow rate command signal **I5** ($\leq I_{min}$) on the basis of the first reference characteristics so that the tilt angle of the swash plate **22R** becomes the minimum tilt angle, and outputs the set flow rate command signal **I5** to the right regulator **23R**. Accordingly, the swash plate **22R** of the right hydraulic pump **21R** rotates up to the minimum tilt angle, and the operating fluid is dispensed from the right hydraulic pump **21R** at the minimum flow rate Q_{min} . Subsequently, when the entire amount of the operating fluid is supplied to the turning hydraulic motor **12** via the straight travel valve **30** and the turning directional control valve **32**, the processing transitions to Step **S22**.

(76) In Step **S22**, which is the turning speed detection step, the speed of turning of the turning body **6** is detected as in Step **S3** and the like. Specifically, the control unit **50A** detects the speed of turning of the turning body **6** on the basis of the signal output from the gyroscope sensor **60**, and when the speed of turning of the turning body **6** is calculated, the processing transitions to Step **S23**. Furthermore, in Step **S23**, which is the turning flow rate calculation step, the turning flow rate of the turning hydraulic motor **12** at the time of turning is calculated as in Step **S4** and the like. Specifically, the control unit **50A** calculates the turning flow rate on the basis of the swept volume of the turning hydraulic motor **12** and the speed reduction ratio between the turning hydraulic motor **12** and the turning body **6**, which are stored in advance, and the speed of turning calculated in Step **S22**, and when the turning flow rate is calculated, the processing transitions to Step **S24**.

(77) In Step **S24**, which is a first pump minimum flow rate calculation step, the minimum flow rate Q_{min} of the right hydraulic pump **21R** is calculated. Specifically, as in Step **S5** and the like, the control unit **50A** calculates the minimum flow rate Q_{min} of the right hydraulic pump **21R** on the basis of the turning flow rate calculated in Step **S23**, but, first, for this purpose, detects the dispense pressure of the right hydraulic pump **21R** on the basis of the signal from the right pressure sensor **62R**. Subsequently, the control unit **50A** calculates the motor leakage amount of the turning hydraulic motor **12** on the basis of the detected dispense pressure of the right hydraulic pump **21R** and the motor efficiency characteristics of the turning hydraulic motor **12**. Lastly, the control unit **50A** calculates the minimum flow rate Q_{min} by adding the calculated motor leakage amount to the turning flow rate. When the minimum flow rate Q_{min} is calculated, the processing transitions to Step **S25**.

(78) In Step **S25**, which is the second supply state switching step, the state of the hydraulic drive system **1A** is switched to the second supply state in which the operating fluid dispensed from the left hydraulic pump **21L**, which is the second hydraulic pump, is supplied to the turning hydraulic motor **12**. Specifically, the control unit **50A** closes the left tank passage **46L** by the left unloader valve **45L** and at the same time, closes the right tank passage **46R** by the right unloader valve **45R**.

At the same time, the control unit **50A** places the spool **30a** of the straight travel valve **30** in the second position **A2**. When the state of the hydraulic supply device **24A** is switched to the second supply state in this manner, the processing transitions to Step **S26**.

(79) In Step **S26**, which is the command electric current setting step, the predetermined flow rate command signal **13** which is set on the basis of the flow rate characteristics stored in advance is output to the left regulator **23L**, as in Step **S9**. The swash plate **22L** of the left hydraulic pump **21L** rotates through a tilt angle corresponding to the flow rate command signal **13**, and the operating fluid is dispensed from the left hydraulic pump **21L** at a flow rate corresponding to the flow rate command signal **13**. Subsequently, the operating fluid is supplied to the turning hydraulic motor **12** via the straight travel valve **30** and the turning directional control valve **32**. Furthermore, the control unit **50A** outputs the flow rate command signal **15** to the right regulator **23R** and causes the right hydraulic pump **21R** to dispense the operating fluid at the dispense flow rate calculated in Step **S24**, namely, the minimum flow rate Q_{min} . The operating fluid dispensed from the right hydraulic pump **21R** in this manner is guided to the right supply passage **34R** via the bypass passage **40R** and the replenishing passage **42** because the right tank passage **46R** is closed; in the right supply passage **34R**, the operating fluid dispensed from the right hydraulic pump **21R** merges with the operating fluid dispensed from the left hydraulic pump **21L** and is supplied to the turning hydraulic motor **12** together with the operating fluid dispensed from the left hydraulic pump **21L**. When the operating fluid after the merging is supplied to the turning hydraulic motor **12** via the straight travel valve **30** and the turning directional control valve **32**, the processing transitions to Step **S27**.

(80) In Step **S27**, which is the turning speed detection step, the speed of turning of the turning body **6** is detected as in Step **S9**. Specifically, the control unit **50A** detects the speed of turning of the turning body **6** on the basis of the signal output from the gyroscope sensor **60**, and when the speed of turning of the turning body **6** is detected, the processing transitions to Step **S28**. Furthermore, in Step **S28**, which is the turning flow rate calculation step, the turning flow rate of the turning hydraulic motor **12** at the time of turning is calculated as in Step **S11** and the like. Specifically, the control unit **50A** calculates the turning flow rate on the basis of the swept volume of the turning hydraulic motor **12** and the speed reduction ratio between the turning hydraulic motor **12** and the turning body **6**, which are stored in advance, and the speed of turning calculated in Step **S27**, and when the turning flow rate is calculated, the processing transitions to Step **S29**.

(81) In Step **S29**, which is the second calibration point obtainment step, an actual dispense flow rate of the left hydraulic pump **21L** is calculated, and a calibration point for the left hydraulic pump **21L** is obtained on the basis of the calculated actual dispense flow rate. Specifically, the control unit **50A** calculates the dispense flow rate of the left hydraulic pump **21L** on the basis of the turning flow rate calculated in Step **S28**, but, first, for this purpose, detects the dispense pressure of the left hydraulic pump **21L** on the basis of the signal from the left pressure sensor **62L**. Subsequently, the control unit **50A** calculates the motor leakage amount of the turning hydraulic motor **12** on the basis of the detected dispense pressure of the left hydraulic pump **21L** and the motor efficiency characteristics of the turning hydraulic motor **12**. Subsequently, the calculated motor leakage amount is added to the turning flow rate to calculate the dispense flow rate; the dispense flow rate calculated in this manner is a total sum of the dispense flow rates of the two hydraulic pumps **21L**, **21R**, namely, a total flow rate. Thus, in order to calculate the dispense flow rate of the left hydraulic pump **21L**, the dispense flow rate of the right hydraulic pump **21R** is subtracted from the total flow rate. Specifically, in Step **S26**, the flow rate command signal **15** is output to the right regulator **23R** so that the right hydraulic pump **21R** dispenses the operating fluid at a predetermined dispense flow rate, that is, the minimum flow rate Q_{min} , meaning that the dispense flow rate of the right hydraulic pump **21R** is known from Step **S24**. Therefore, the control unit **50A** calculates the dispense flow rate of the left hydraulic pump **21L** (=the turning flow rate+the motor leakage amount−the minimum flow rate Q_{min}) by subtracting the known dispense flow rate, that is, the

minimum flow rate Q_{min} (correction flow rate), from the total flow rate. When the dispense flow rate of the left hydraulic pump **21L** is calculated, the control unit **50A** stores the calculated dispense flow rate in association with the flow rate command signal **I3** set in Step **S26**, meaning that the control unit **50A** obtains the calibration point **73** (refer to FIG. 3). When the first calibration point, namely, the calibration point **73**, is obtained in this manner, the processing transitions to Step **S30**. (82) In Step **S30**, which is the number-of-calibration-points checking step, whether or not two or more calibration points have been obtained is determined at the time of calibrating the second reference characteristics. Note that the number of calibration points to be obtained may be three or more. When the number of calibration points obtained is determined as one, the processing returns to Step **S26**, the flow rate command signal **I4** is output to the left regulator **23L**, then the speed of turning is detected (Step **S27**), and furthermore, the turning flow rate is calculated on the basis of the speed of turning detected in Step **27** (Step **S28**). Furthermore, the control unit **50A** calculates the dispense flow rate on the basis of the turning flow rate detected in Step **S28** and stores the calculated flow rate and the flow rate command signal **I4** in association with each other (Step **S29**). When the second calibration point, namely, the calibration point **74**, is obtained in this manner (refer to FIG. 3), the processing transitions from Step **S30** to Step **S31**.

(83) In Step **S31**, which is the second pump flow rate calibration step, the second reference characteristics are calibrated on the basis of the two calibration points **73**, **74** obtained in Step **S29**, as in Step **S14** according to Embodiment 1. Specifically, in the range where the flow rate Q satisfies the relationship: $Q_{min} \leq Q \leq Q_{max}$, a straight line passing through the two calibration points **73**, **74** (refer to the double-dot-dashed line in FIG. 3) is calculated as the second actual measurement characteristics, which are actual flow rate characteristics of the left hydraulic pump **21L**. More specifically, the control unit **50A** calculates, on the basis of the two calibration points **73**, **74**, a slope and an intercept of the second actual measurement characteristics in the range $Q_{min} \leq Q \leq Q_{max}$, calculates the second actual measurement characteristics, and sets the calculated second actual measurement characteristics as new second reference characteristics. When the second reference characteristics are calibrated on the basis of the second actual measurement characteristics in this manner, the second pump calibration process ends, and the flow rate calibration process also ends.

(84) Thus, in the hydraulic drive system **1A**, by performing the aforementioned flow rate calibration process, it is possible to more accurately calibrate the flow rate characteristics of the two hydraulic pumps **21L**, **21R** in the case where the replenishing unit **47** is provided. Therefore, in the excavator **3** with the hydraulic drive system **1A** mounted thereon, the dispense flow rates of the two hydraulic pumps **21L**, **21R** can be controlled with high accuracy.

(85) Aside from this, the hydraulic drive system **1A** according to Embodiment 2 produces substantially the same advantageous effects as the hydraulic drive system **1** according to Embodiment 1.

Embodiment 3

(86) A hydraulic drive system **1B** according to Embodiment 3 has the same configuration as the hydraulic drive system **1A** according to Embodiment 2, as illustrated in FIG. 5. However, the second pump calibration process in the flow rate calibration process which is performed by a control unit **50B** in the hydraulic drive system **1B** is different from that performed by the control unit **50A** in the hydraulic drive system **1A** according to Embodiment 2. Hereinafter, the second pump calibration process which is performed by the control unit **50B** will be described in detail. Specifically, when the control unit **50B** performs Steps **S1** to **S7** of the flow rate calibration process as illustrated in FIG. 6 and the calibration of the flow rate characteristics of the right hydraulic pump **21R**, in other words, the calibration of the first reference characteristics, ends, the control unit **50B** causes the processing to transition to Step **S40**, performs the second pump calibration process such as that illustrated in FIG. 8, and causes the processing to transition to Step **S41**.

(87) In Step **S41**, which is the second supply state switching step, the state of the hydraulic drive

system **1B** is switched to the second supply state in which the operating fluid dispensed from the left hydraulic pump **21L**, which is the second hydraulic pump, is supplied to the turning hydraulic motor **12**. Specifically, the control unit **50B** completely opens the right tank passage **46R** by the right unloader valve **45R**, which is one example of the exhaust valve, and closes the left tank passage **46L** by the left unloader valve **45L**. Furthermore, the control unit **50B** places the spool **30a** of the straight travel valve **30** in the second position **A2** and operates the turning directional control valve **32**, causing the operating fluid in the right hydraulic pump **21R** to be supplied to the turning hydraulic motor **12**. When the state of the hydraulic supply device **24A** is switched to the second supply state, the processing transitions to Step **S42**.

(88) In Step **S42**, which is the command electric current setting step, the predetermined flow rate command signal **I3** which is set on the basis of the flow rate characteristics stored in advance is output to the left regulator **23L**, as in Step **S26**. The swash plate **22L** of the left hydraulic pump **21L** rotates through a tilt angle corresponding to the flow rate command signal **I3**, and the operating fluid is dispensed from the left hydraulic pump **21L** at a flow rate corresponding to the flow rate command signal **I3**. Subsequently, when the operating fluid is supplied to the turning hydraulic motor **12** via the straight travel valve **30** and the turning directional control valve **32**, the processing transitions to Step **S43**. In Step **S43**, which is the turning speed detection step, the speed of turning of the turning body **6** is detected as in Step **S27**. Specifically, the control unit **50B** detects the speed of turning of the turning body **6** on the basis of the signal output from the gyroscope sensor **60**, and when the speed of turning of the turning body **6** is detected, the processing transitions to Step **S44**. Furthermore, in Step **S44**, which is the turning flow rate calculation step, the turning flow rate of the turning hydraulic motor **12** at the time of turning is calculated as in Step **S28**. Specifically, the control unit **50B** calculates the turning flow rate on the basis of the swept volume of the turning hydraulic motor **12** and the speed reduction ratio between the turning hydraulic motor **12** and the turning body **6**, which are stored in advance, and the speed of turning calculated in Step **S43**, and when the turning flow rate is calculated, the processing transitions to Step **S45**.

(89) In Step **S45**, which is the second calibration point obtainment step, an actual dispense flow rate of the left hydraulic pump **21L** is calculated, and a calibration point for the left hydraulic pump **21L** is obtained on the basis of the calculated actual dispense flow rate. Specifically, the control unit **50B** calculates the dispense flow rate of the left hydraulic pump **21L** on the basis of the turning flow rate calculated in Step **S45**, but, first, for this purpose, detects the dispense pressure of the left hydraulic pump **21L** on the basis of the signal from the left pressure sensor **62L**. Subsequently, the control unit **50B** calculates the motor leakage amount of the turning hydraulic motor **12** on the basis of the detected dispense pressure of the left hydraulic pump **21L** and the motor efficiency characteristics of the turning hydraulic motor **12**. Furthermore, the control unit **50B** calculates the dispense flow rate of the left hydraulic pump **21L** on the basis of the calculated motor leakage amount and turning flow rate.

(90) Specifically, in the hydraulic drive system **1B**, the replenishing unit **47** is provided, and the right tank passage **46R** is completely open. Therefore, a portion of the operating fluid dispensed from the left hydraulic pump **21L** flows to the tank **27** via the replenishing unit **47**, the right pump passage **33R**, and the tank passage **46R**; the control unit **50B** calculates an outflow rate Q_a , which is the flow rate of the operating fluid flowing to the tank **27**, in addition to the motor leakage amount. More specifically, the control unit **50B** detects the dispense pressure of the right hydraulic pump **21R** on the basis of the signal from the right pressure sensor **62R** (first pressure sensor), and calculates the outflow rate Q_a on the basis of the detected dispense pressure and the dispense pressure detected by the left pressure sensor **62L** (second pressure sensor). In other words, the control unit **50B** calculates the outflow rate Q_a on the basis of the following Expression 1.

(91)

Expression 1

$$Qa = C \times d \times \sqrt{2(P2 - P1)}$$

Expression 1

(92) Here, C is a flow rate coefficient, d is a throttle diameter that is the diameter of the throttle **47b**, P1 is the dispense pressure of the right hydraulic pump **21R**, P2 is the dispense pressure of the left hydraulic pump **21L**, ρ is the liquid density of the operating fluid; the flow rate coefficient C, the throttle diameter d, and the liquid density ρ are stored by the control unit **50B** in advance. When the control unit **50B** detects the two dispense pressures P1, P2, the control unit **50B** calculates the outflow rate Qa on the basis of these dispense pressures and Expression 1. In other words, the control unit **50B** constitutes an outflow rate detection device together with the two pressure sensors **62L**, **62R** and calculates the outflow rate on the basis of the dispense flow rates P1, P2 detected on the basis of the signals from the two pressure sensors **62L**, **62R**. Subsequently, the control unit **50B** calculates the dispense flow rate of the left hydraulic pump **21L** by adding the calculated motor leakage amount and outflow rate Qa to the turning flow rate. When the dispense flow rate of the left hydraulic pump **21L** is calculated, the control unit **50B** stores the calculated dispense flow rate in association with the flow rate command signal I3 set in Step **S42**, meaning that the control unit **50B** obtains the calibration point **73** (refer to FIG. 3). When the first calibration point, namely, the calibration point **73**, is obtained in this manner, the processing transitions to Step **S46**.

(93) In Step **S46**, which is the number-of-calibration-points checking step, whether or not two or more calibration points have been obtained is determined at the time of calibrating the second reference characteristics. Note that the number of calibration points to be obtained may be three or more. When the number of calibration points obtained is determined as one, the processing returns to Step **S42**, the flow rate command signal I4 is output to the left regulator **23L**, then the speed of turning is detected (Step **S43**), and furthermore, the turning flow rate is calculated on the basis of the speed of turning detected in Step **43** (Step **S44**). Furthermore, the control unit **50B** calculates the dispense flow rate on the basis of the turning flow rate detected in Step **S44** and stores the calculated flow rate and the flow rate command signal I4 in association with each other (Step **S45**). When the second calibration point, namely, the calibration point **74**, is obtained in this manner (refer to FIG. 3), the processing transitions from Step **S46** to Step **S47**.

(94) In Step **S47**, which is the second pump flow rate calibration step, the second reference characteristics are calibrated on the basis of the two calibration points **73**, **74** obtained in Step **S45**, as in Step **S14** according to Embodiment 1. Specifically, in the range where the flow rate Q satisfies the relationship: $Q_{min} \leq Q \leq Q_{max}$, a straight line passing through the two calibration points **73**, **74** (refer to the double-dot-dashed line in FIG. 3) is calculated as the second actual measurement characteristics, which are actual flow rate characteristics of the left hydraulic pump **21L**. More specifically, the control unit **50B** calculates, on the basis of the two calibration points **73**, **74**, a slope and an intercept of the second actual measurement characteristics in the range $Q_{min} \leq Q \leq Q_{max}$, calculates the second actual measurement characteristics, and sets the calculated second actual measurement characteristics as new second reference characteristics. When the

second reference characteristics are calibrated on the basis of the second actual measurement characteristics in this manner, the second pump calibration process ends, and the flow rate calibration process also ends.

(95) Thus, in the hydraulic drive system **1B**, by performing the flow rate calibration process having a different flow of steps compared to the flow of steps for the hydraulic drive system **1A** according to Embodiment 2, it is possible to more accurately calibrate the flow rate characteristics of the two hydraulic pumps **21L**, **21R**, as in the case of the hydraulic drive system **1A**. Therefore, in the excavator **3** with the hydraulic drive system **1B** mounted thereon, the dispense flow rates of the two hydraulic pumps **21L**, **21R** can be controlled with high accuracy.

(96) Aside from this, the hydraulic drive system **1B** according to Embodiment 3 produces substantially the same advantageous effects as the hydraulic drive system **1A** according to Embodiment 2.

Embodiment 4

(97) A hydraulic drive system **1C** according to Embodiment 4 has the same configuration as the hydraulic drive system **1A** according to Embodiment 2, as illustrated in FIG. 5. However, the second pump calibration process in the flow rate calibration process which is performed by a control unit **50C** in the hydraulic drive system **1C** is completely different from those performed in the hydraulic drive system **1A** according to Embodiment 2 and the hydraulic drive system **1B** according to Embodiment 3. Hereinafter, the second pump calibration process which is performed by the control unit **50C** will be described. Specifically, when the control unit **50C** performs Steps **S1** to **S5** of the flow rate calibration process as illustrated in FIG. 6 and the calibration of the flow rate characteristics of the right hydraulic pump **21R** ends, the control unit **50C** causes the processing to transition to Step **S50**, performs the second pump calibration process such as that illustrated in FIG. 9, and causes the processing to transition to Step **S51**. The processing transitions to Step **S51**.

(98) In Step **S51**, which is a third supply state switching step, the state of the hydraulic drive system **1C** is switched to a third supply state in which the operating fluid dispensed from the two hydraulic pumps **21L**, **21R** is supplied to the turning hydraulic motor **12**. Specifically, the control unit **50C** outputs signals to the valves **30**, **31L**, **31R**, **32**, **45L**, **45R**, and controls the operation thereof in the following manner. More specifically, the control unit **50C** closes the left tank passage **46L** by the left unloader valve **45L** and closes the right tank passage **46R** by the right unloader valve **45R**. Furthermore, the control unit **50C** moves the spool **30a** of the straight travel valve **30** to the merging function and causes the operating fluid dispensed from the two hydraulic pumps **21L**, **21R** to merge at the straight travel valve **30** so that the operating fluid is guided to the right supply passage **34R**.

(99) The control unit **50C** operates the turning directional control valve **32**, in other words, causes the spool **32a** of the turning directional control valve **32** to slide. Accordingly, the operating fluid guided to the right supply passage **34R** is supplied to the turning hydraulic motor **12**. At this time, the spool **32a** is slid so that the degree of opening of the turning directional control valve **32** reaches the maximum degree. On the other hand, the control unit **50C** places each of the spools **31La**, **31Ra** (including the spools of various directional control valves) of the directional control valves **31L**, **31R** (including various directional control valves corresponding to the boom cylinder **13**, the arm cylinder **14**, the bucket cylinder **15**, and the like) other than the turning directional control valve **32** in the neutral position, thereby preventing the operating fluid from flowing to the other hydraulic actuators such as the left traveling hydraulic motor **11L** (the second hydraulic actuator) and the right traveling hydraulic motor **11R**. In this manner, only the spool **32a** of the turning directional control valve **32** is slid to cause the entire operating fluid in the two hydraulic pumps **21L**, **21R** to be supplied to the turning hydraulic motor **12** alone. When the state of the hydraulic supply device **24A** is switched to the third supply state in which the entire operating fluid in the two hydraulic pumps **21L**, **21R** is supplied to the turning hydraulic motor **12** alone in this

manner, the processing transitions to Step S52.

(100) In Step S52, which is the command electric current setting step, the predetermined flow rate command signal I3 which is set on the basis of the flow rate characteristics stored in advance is output to the left regulator 23L, as in Steps S26, S42. The swash plate 22L of the left hydraulic pump 21L rotates through a tilt angle corresponding to the flow rate command signal I3, and the operating fluid is dispensed from the left hydraulic pump 21L at a flow rate corresponding to the flow rate command signal I3. Meanwhile, a predetermined flow rate command signal that is the flow rate command signal I5 ($\leq I_{min}$) in the present embodiment is output to the right regulator 23R as well. The swash plate 22L of the left hydraulic pump 21L rotates up to the minimum tilt angle, meaning that the dispense flow rate of the left hydraulic pump 21L is set to the minimum flow rate Q_{min} . The entire amount of the operating fluid dispensed from the two hydraulic pumps 21L, 21R in this manner is supplied to the turning hydraulic motor 12 via the straight travel valve 30 and the turning directional control valve 32. When the operating fluid is supplied in this manner, the processing transitions to Step S53.

(101) In Step S53, which is the turning speed detection step, the speed of turning of the turning body 6 is detected as in Step S3 and the like. Specifically, the control unit 50C detects the speed of turning of the turning body 6 on the basis of the signal output from the gyroscope sensor 60, and when the speed of turning of the turning body 6 is calculated, the processing transitions to Step S54. Furthermore in Step S54, which is the turning flow rate calculation step, the turning flow rate of the turning hydraulic motor 12 at the time of turning is calculated as in Step S4 and the like. Specifically, the control unit 50C calculates the turning flow rate on the basis of the swept volume of the turning hydraulic motor 12 and the speed reduction ratio between the turning hydraulic motor 12 and the turning body 6, which are stored in advance, and the speed of turning calculated in Step S53, and when the turning flow rate is calculated, the processing transitions to Step S55.

(102) In Step S55, which is the second calibration point obtainment step, an actual dispense flow rate of the left hydraulic pump 21L is calculated, and a calibration point for the left hydraulic pump 21L is obtained on the basis of the calculated actual dispense flow rate. Specifically, the control unit 50C calculates the dispense flow rate of the left hydraulic pump 21L on the basis of the turning flow rate calculated in Step S54, but, first, for this purpose, detects at least one of the dispense pressures of the two hydraulic pumps 21L, 21R on the basis of the signals from the pressure sensors 62L, 62R. Subsequently, the control unit 50C calculates the motor leakage amount of the turning hydraulic motor 12 on the basis of the detected dispense pressure and the motor efficiency characteristics of the turning hydraulic motor 12. Subsequently, the calculated motor leakage amount is added to the turning flow rate to calculate the dispense flow rate; the dispense flow rate calculated in this manner is a total sum of the dispense flow rates of the two hydraulic pumps 21L, 21R, namely, a total flow rate. Thus, in order to calculate the dispense flow rate of the left hydraulic pump 21L, the dispense flow rate of the right hydraulic pump 21R is subtracted from the total flow rate.

(103) Specifically, in Step S55, the flow rate command signal I5 is output to the right regulator 23R so that the right hydraulic pump 21R dispenses the operating fluid at a predetermined dispense flow rate, that is, the minimum flow rate Q_{min} . The flow rate characteristics of the right hydraulic pump 21R, namely, the first reference characteristics, have already been calibrated in Step S7, and the dispense flow rate of the right hydraulic pump 21R can be calculated on the basis of the first reference characteristics and the flow rate command signal I5. Therefore, the control unit 50C calculates the dispense flow rate of the left hydraulic pump 21L (=the turning flow rate+the motor leakage amount−the minimum flow rate Q_{min}) by subtracting the calculated dispense flow rate, that is, the minimum flow rate Q_{min} (correction flow rate), from the total flow rate. When the dispense flow rate of the left hydraulic pump 21L is calculated, the control unit 50C stores the calculated dispense flow rate in association with the flow rate command signal I3 set in Step S52, meaning that the control unit 50C obtains the calibration point 73 (refer to FIG. 3). When the first

calibration point, namely, the calibration point **74**, is obtained in this manner, the processing transitions to Step **S56**.

(104) In Step **S56**, which is the number-of-calibration-points checking step, whether or not two or more calibration points have been obtained is determined at the time of calibrating the second reference characteristics, as in Step **S30** according to Embodiment 2. Note that the number of calibration points to be obtained may be three or more. When the number of calibration points obtained is determined as one, the processing returns to Step **S52**, the flow rate command signal **I4** is output to the left regulator **23L**, then the speed of turning is detected (Step **S53**), and furthermore, the turning flow rate is calculated on the basis of the speed of turning detected in Step **S3** (Step **S54**). Furthermore, the control unit **50C** calculates the dispense flow rate on the basis of the turning flow rate detected in Step **S54** and stores the calculated flow rate and the flow rate command signal **I4** in association with each other (Step **S55**). When the second calibration point, namely, the calibration point **74**, is obtained in this manner (refer to FIG. 3), the processing transitions from Step **S56** to Step **S57**.

(105) In Step **S57**, which is the second pump flow rate calibration step, the second reference characteristics are calibrated on the basis of the two calibration points **73**, **74** obtained in Step **S55**, as in Step **S31** according to Embodiment 2. Specifically, the control unit **50C** calculates the second actual measurement characteristics on the basis of the two calibration points **73**, **74**, and the calculated second actual measurement characteristics are set as new second reference characteristics. When the second reference characteristics are calibrated on the basis of the second actual measurement characteristics in this manner, the second pump calibration process ends, and the flow rate calibration process also ends.

(106) Thus, in the hydraulic drive system **1C**, by performing the aforementioned flow rate calibration process, it is possible to more accurately calibrate the flow rate characteristics of the two hydraulic pumps **21L**, **21R** in the case where the replenishing unit **47** is provided. Therefore, in the excavator **3** with the hydraulic drive system **1C** mounted thereon, the dispense flow rates of the two hydraulic pumps **21L**, **21R** can be controlled with high accuracy.

Embodiment 5

(107) The pump flow rate calibration system may be a hydraulic drive system **1D** according to Embodiment 5 to be described below. Specifically, the hydraulic drive system **1D** according to Embodiment 5 is a system that drives a hydraulic motor **12D** by supplying the operating fluid thereto, as illustrated in FIG. 10, and includes a hydraulic pump **21D**, a regulator **23D**, and a hydraulic supply device **24D**. The hydraulic pump **21D** is what is called a variable-capacitance swash plate pump and includes a swash plate **22D**. The hydraulic pump **21D** is capable of changing a dispense flow rate thereof by rotating the swash plate **22D**, and a regulator **23D** is provided on the hydraulic pump **21D** in order to rotate the swash plate **22D**. The regulator **23D** adjusts, according to the flow rate command signal input thereto, the tilt angle of the swash plate **22D** and controls the dispense flow rate of the hydraulic pump **21D**. The hydraulic supply device **24D** is connected to the hydraulic pump **21D** configured as just described, in order to supply the dispensed operating fluid to the hydraulic motor **12D**.

(108) The hydraulic supply device **24D** includes a directional control valve **32D** and can control the flow and the flow rate of the operating fluid that is supplied to the hydraulic motor **12D**. More specifically, the directional control valve **32D** is connected to the hydraulic motor **12D** and the tank **27** in addition to the hydraulic pump **21D** and can switch the connection between the hydraulic motor **12D** and each of the hydraulic pump **21D** and the tank **27**. In other words, the directional control valve **32D** includes a spool **32Da** and switches said connection by changing the position of the spool **32Da**. The spool **32Da** receives pilot pressures output from two different electromagnetic proportional control valves **32Db**, **32Dc** provided at both ends of the spool **32Da** and moves from a neutral position in either of the opposite directions in accordance with the difference between the two pilot pressures received. Accordingly, the connection between the hydraulic motor **12D** and

each of the hydraulic pump **21D** and the tank **27** can be switched, and by switching the connection and changing the flow direction of the operating fluid, it is possible to change the direction of rotation of the hydraulic motor **12D**. Furthermore, the spool **32Da** moves to a position corresponding to the difference between two pilot pressures, and the degree of opening of the directional control valve **32D** is thereby adjusted to reach a degree of opening corresponding to said position.

(109) Note that the following elements are connected between the directional control valve **32D** and the hydraulic motor **12D**. Specifically, the directional control valve **32D** is connected to the hydraulic motor **12** via two turning supply passages **37DL**, **37DR**, and relief valves **38DL**, **38DR** are connected to the two turning supply passages **37DL**, **37DR**, respectively. When the hydraulic pressure of the operating fluid flowing through the turning supply passages **37DL**, **37DR** connected to the two relief valves **38DL**, **38DR** exceeds a predetermined relief pressure, the two relief valves **38DL**, **38DR** discharge the operating fluid to the tank **27**. Furthermore, the two turning supply passages **37DL**, **37DR** are connected to the tank **27** via check valves **39DL**, **39DR** and are designed to be able to add the operating fluid from the tank **27** when there is a shortage of the operating fluid.

(110) The hydraulic drive system **1D** configured as described above further includes a control unit **50D**, and the operation of the regulator **23D** and the directional control valves **32D** is controlled by the control unit **50D**. Furthermore, an operation device **51D** is electrically connected to the control unit **50D** in order to provide a command related to the operation of the hydraulic supply device **24D**. The operation device **51D** includes, for example, an electric joystick or a remote control valve. Specifically, the operation device **51D** includes an operation lever **51Da**; when the operation lever **51Da** is pulled down, the operation device **51D** outputs, to the control unit **50D**, a signal corresponding to the extent of how much the operation lever **51Da** is pulled down.

(111) The control unit **50D** is designed to control the operation of the directional control valve **32D** in accordance with the signal output from the operation device **51D**; the control unit **50D** is configured as follows in order to control the operation of the directional control valve **32D**. Specifically, the control unit **50D** is electrically connected to the electromagnetic proportional control valves **32Db**, **32Dc** provided on the directional control valve **32D** and outputs command signals to the electromagnetic proportional control valves **32Db**, **32Dc** in accordance with the signal output from the operation device **51D**. Thus, the electromagnetic proportional control valves **32Db**, **32Dc** output pilot pressures corresponding to the command signals, and the spool **32Da** moves to a position corresponding to the difference between the two pilot pressures. Accordingly, the directional control valve **32** opens with a degree of opening corresponding to the amount of operation on the operation lever **51Da**, and the operating fluid is guided to the hydraulic motor **12D** at a flow rate corresponding to the amount of the operation on the operation lever **51Da**.

(112) Furthermore, the hydraulic drive system **1D** includes a rotation sensor **60D** and a pressure sensor **62D**. The rotation sensor **60D** is provided on an output shaft **12a** of the hydraulic motor **12D** and is electrically connected to the control unit **50D**. Furthermore, the rotation sensor **60D** outputs, to the control unit **50D**, a signal corresponding to the speed of rotation of the output shaft **12a**, and the control unit **50D** detects the speed of rotation of the hydraulic motor **12D** on the basis of a signal from the rotation sensor **60D**. Moreover, the pressure sensor **62D** is connected to the hydraulic pump **21D** and is electrically connected to the control unit **50D**. The pressure sensor **62D** disposed as just described outputs, to the control unit **50D**, a signal corresponding to the dispense pressure of the hydraulic pump **21D**, and the control unit **50D** detects the dispense pressure of the hydraulic pump **21D** on the basis of the signal output from the pressure sensor **62D**. In addition, the control unit **50D** performs various calculations and stores a variety of information.

(113) In the hydraulic drive system **1D** configured as described above, the control unit **50D** controls the operation of the hydraulic supply device **24D** in accordance with the operation performed on the operation device **51D** and operates a hydraulic actuator **12D**. Specifically, when

the operation lever **51Da** is operated and a signal is output from the operation device **51D**, the control unit **50D** outputs, to the electromagnetic proportional control valve **32Db** (or the electromagnetic proportional control valve **32Dc**), a rotation command signal corresponding to said signal, and operates the directional control valve **32D**. Accordingly, the operating fluid is supplied from the hydraulic pump **21D** to the hydraulic motor **12D**, and the hydraulic motor **12D** rotates with the operating fluid supplied thereto. Furthermore, the control unit **50D** causes the directional control valve **32D** to open with a degree of opening corresponding to the amount of the operation on the operation lever **51Da**, and controls the dispense flow rate of the hydraulic pump **21D** via the regulator **23D** in accordance with the amount of the operation on the operation lever **51Da**. Thus, it is possible to rotate the hydraulic motor **12D** at a speed of rotation corresponding to the amount of the operation on the operation lever **51Da**.

(114) The control unit **50D** having such functions sets reference characteristics for the hydraulic pump **21D** in advance and calibrates the set flow rate characteristics, as with the control units **50**, **50A**, **50B** according to Embodiments 1 to 3. Hereinafter, the hydraulic-pump flow-rate calibration process which is performed by the control unit **50D** will be described. Specifically, the control unit **50D** determines whether or not a predetermined calibration condition is met, and when the calibration condition is met, performs a flow rate calibration process such as that illustrated in FIG. **11**. When the flow rate calibration process is performed, the processing transitions to Step **S61**.

(115) In Step **S61**, which is a supply state switching step, the state of the hydraulic drive system **1D** is switched to a supply state in which the operating fluid dispensed from the hydraulic pump **21D** is supplied to the hydraulic motor **12D**. Specifically, the control unit **50D** outputs a signal to the electromagnetic proportional control valve **32Db** (or the electromagnetic proportional control valve **32Dc**) of the directional control valve **32D**, operates the spool **32Da** of the directional control valve **32D**, and connects the hydraulic pump **21D** and the tank **27** to the hydraulic motor **12D**. At this time, in order that the entire amount of the operating fluid in the hydraulic pump **21D** shall be supplied to the hydraulic motor **12D**, the spool **32Da** is slid so that the degree of opening of the directional control valve **32D** reaches the maximum degree. When the spool **32Da** is slid in this manner and the state of the hydraulic supply device **24D** is switched to the supply state, the processing transitions to Step **S62**.

(116) In Step **S62**, which is the command electric current setting step, the predetermined flow rate command signal **I1** which is set on the basis of the reference characteristics is output to the regulator **23D**, as in Step **S2** described above. Accordingly, the swash plate **22D** of the hydraulic pump **21D** rotates through a tilt angle corresponding to the flow rate command signal **I1**, and the operating fluid is dispensed from the hydraulic pump **21D** at a flow rate corresponding to the flow rate command signal **I1**. Subsequently, when the entire amount of the operating fluid is supplied to the hydraulic motor **12D** via the directional control valve **32D**, the processing transitions to Step **S63**. In Step **S63**, which is a rotation speed detection step, the speed of rotation of the hydraulic motor **12D** is detected. Specifically, the control unit **50D** detects the speed of rotation of the hydraulic motor **12D** on the basis of the signal output from the rotation sensor **60D**. Subsequently, when the speed of rotation of the hydraulic motor **12D** is detected, the processing transitions to Step **S64**.

(117) In Step **S64**, which is a supply flow rate calculation step, the flow rate of the operating fluid supplied to the hydraulic motor **12D** during rotation of the hydraulic motor **12D**, namely, a supply flow rate, is calculated. Specifically, the control unit **50D** stores the swept volume of the hydraulic motor **12D** in advance and calculates the supply flow rate on the basis of said swept volume and the speed of rotation detected in Step **S63**. More specifically, the supply flow rate is calculated by multiplying the speed of rotation detected in Step **S63** by the swept volume. When the supply flow rate is calculated, the processing transitions to Step **S65**.

(118) In Step **S65**, which is a calibration point obtainment step, the actual dispense flow rate of the hydraulic pump **21D** is calculated, and a calibration point for the hydraulic pump **21** is obtained on

the basis of the calculated actual dispense flow rate. Specifically, the control unit **50D** calculates the dispense flow rate of the hydraulic pump **21D** on the basis of the supply flow rate calculated in Step **S64**, but, first, for this purpose, detects the dispense pressure of the hydraulic pump **21D** on the basis of the signal from the pressure sensor **62D**. Subsequently, the control unit **50D** calculates the motor leakage amount of the hydraulic motor **12D** on the basis of the detected dispense pressure and adds the calculated motor leakage amount to the turning flow rate. Thus, the dispense flow rate (=the turning flow rate+the motor leakage amount) of the hydraulic motor **12D** is calculated. When the control unit **50D** calculates the dispense flow rate, the control unit **50D** stores the calculated dispense flow rate in association with the flow rate command signal **I1** set in Step **S62**. For example, in the case where the dispense flow rate applied in response to the flow rate command signal **I1** is high compared to the reference characteristics (the solid line in FIG. 3), the calibration point **71** is obtained, as illustrated in FIG. 3. When the first calibration point, namely, the calibration point **71**, is calculated in this manner, the processing transitions to Step **S66**. (119) In Step **S66**, which is the number-of-calibration-points checking step, whether or not two or more calibration points have been obtained is determined at the time of calibrating the reference characteristics. Note that the number of calibration points to be obtained may be three or more. When the number of calibration points obtained is determined as one, the processing returns to Step **S62**, the flow rate command signal **I2** is output to the regulator **23D**, then the speed of turning is detected (Step **S63**), and furthermore, the turning flow rate is calculated on the basis of the speed of turning detected in Step **63** (Step **S64**). Furthermore, the control unit **50D** calculates the dispense flow rate on the basis of the turning flow rate detected in Step **S64** and stores the calculated flow rate and the flow rate command signal **I2** in association with each other (Step **S65**). When the second calibration point, namely, the calibration point **74**, is obtained in this manner (refer to FIG. 3), the processing transitions to Step **S67**.

(120) In Step **S67**, which is a pump flow rate calibration step, the reference characteristics are calibrated on the basis of the two calibration points **71**, **72** obtained in Step **S65**, as in Step **S14** according to Embodiment 1. Specifically, in the range where the flow rate Q satisfies the relationship: $Q_{min} \leq Q \leq Q_{max}$, a straight line passing through the two calibration points **71**, **72** (refer to the double-dot-dashed line in FIG. 3) is calculated as the actual measurement characteristics, which are actual flow rate characteristics of the hydraulic pump **21D**. More specifically, the control unit **50D** calculates, on the basis of the two calibration points **71**, **72**, a slope and an intercept of the actual measurement characteristics in the range $Q_{min} \leq Q \leq Q_{max}$, calculates the actual measurement characteristics, and sets the calculated actual measurement characteristics as new reference characteristics. When the reference characteristics are calibrated on the basis of the actual measurement characteristics in this manner, the flow rate calibration process ends.

(121) Thus, in the hydraulic drive system **1D**, by performing the flow rate calibration process such as that described above, it is possible to calibrate the dispense flow rate of the hydraulic pump **21D** in the state where the hydraulic drive system **1D** includes the hydraulic pump **21D**. This means that in the hydraulic drive system **1D**, the dispense flow rate of the hydraulic pump **21D** can be controlled with high accuracy. Furthermore, the hydraulic drive system **1D** can calculate the dispense flow rate of the hydraulic pump **21D** on the basis of the speed of rotation of the hydraulic motor detected by the rotation sensor **60D** and calibrate the flow rate characteristics on the basis of the calculated dispense flow rate. This means that in the hydraulic drive system **1D**, the flow rate characteristics of the hydraulic pump **21D** can be calibrated without addition of a flow rate sensor, and it is possible to minimize an increase in the number of components for the purpose of calibration.

Other Embodiments

(122) The above description focuses on the case where the hydraulic drive systems **1**, **1A**, **1B** according to Embodiments 1 to 3 are mounted on the excavator **3**, but the excavator **3** is not

necessarily the only option and may be replaced by other construction equipment such as a crane and a wheel loader. Furthermore, the construction equipment is not necessarily the only option; the hydraulic drive system may be applied to a robot of the hydraulic drive type, and in this case, water such as saline may be used as the operating fluid.

(123) Note that in the case of a crane, the hydraulic-pump flow-rate calibration process may be performed using a hoist motor provided on a hoist device for the crane instead of a turning motor. In the case of a wheel loader or the like, the hydraulic-pump flow-rate calibration process may be performed using a traveling motor instead of the turning motor. Furthermore, the hydraulic-pump flow-rate calibration process may be performed using a cylinder instead of the hydraulic motor. Specifically, it is sufficient that a supply flow rate for the hydraulic actuator be calculated according to the amount of stroke of the cylinder and the hydraulic-pump flow-rate calibration process be performed on the basis of the calculated supply flow rate. At this time, a stroke sensor functions as the flow rate detection device. Furthermore, the flow rate detection device does not necessarily need to be the gyroscope sensor **60** or the stroke sensor and may be a flowmeter or the like provided in a passage connected to each hydraulic actuator. Moreover, in the hydraulic drive systems **1**, **1A**, **1B** according to Embodiments 1 to 3, the three-axis gyroscope sensor is used as the gyroscope sensor **60**, but a two-axis gyroscope sensor may be used.

(124) Furthermore, in the hydraulic drive systems **1**, **1A** to **1C** according to Embodiments 1 to 4, the traveling directional control valves **31L**, **31R** are configured to operate on the basis of the pilot pressures output from the electromagnetic proportional control valves **31Lb**, **31Lc**, **31Rb**, **31Rc**, but do not necessarily need to have such a configuration. Specifically, the traveling operation device **52** may include a remote control valve of the hydraulic type, and the traveling directional control valves **31L**, **31R** may be directional control valves of the hydraulic drive type that are driven with a pilot pressure output from the remote control valve. In this case, the pressure sensor or the like detects the pilot pressure output from the remote control valve, and thus whether or not the traveling operation device **52** has been operated is detected.

(125) Furthermore, in the hydraulic drive systems **1**, **1A** to **1D** according to Embodiments 1 to 5, the reference characteristics are calibrated on the basis of two or more calibration points, but the number of calibration points do not necessarily need to be two or more. Specifically, a changing point **75** of the minimum flow rate Q_{min} for each of the hydraulic pumps **21L**, **21R**, **21D** varies to a greater extent among products than a changing point **76** of the maximum flow rate Q_{max} for the hydraulic pump and can be regarded as a substantially fixed point. Therefore, the actual measurement characteristics can be calculated on the basis of the changing point **75** and one calculated calibration point, and the reference characteristics can be calibrated on the basis of the calculated actual measurement characteristics. Furthermore, in the case where there is hysteresis in the reference characteristics of the hydraulic pumps **21L**, **21R**, **21D**, two calibration points may be calculated at the times when the flow rate increases and when the flow rate decreases, and the reference characteristics may be calibrated for each of the cases where the flow rate increases and where the flow rate decreases. Moreover, the flow rates included in the reference characteristics for when the tilt angle is minimum and when the tilt angle is maximum, in other words, the minimum flow rate Q_{min} and the maximum flow rate Q_{max} of the hydraulic pumps **21L**, **21R**, **21D**, may be calibrated in the aforementioned method.

(126) Furthermore, the hydraulic drive systems **1**, **1A**, **1B** according to Embodiments 1 to 3 include the unloader valves **45L**, **45R**, but the unloader valves **45L**, **45R** do not necessarily need to be included; a hydraulic drive system **1E** illustrated in FIG. **12** is applicable. Specifically, a bypass cut-off valve **49L** is provided in the left bypass passage **40L** in the hydraulic drive system **1E**, and the left bypass passage **40L** is connected to the tank **27** via the bypass cut-off valve **49L**. Furthermore, a directional control valve (for example, a bucket directional control valve and a first boom directional control valve) not illustrated in the drawings is provided in the left bypass passage **40L**, on the upstream side of the bypass cut-off valve **49L**, but on the downstream side of

the left traveling directional control valve **31L**, and the degree of opening of the left bypass passage **40L** is adjusted according to the position of the spool of each directional control valve including the directional control valve **31L**. Meanwhile, a bypass cut-off valve **49R** is likewise provided in the right bypass passage **40R**, and the right bypass passage **40R** is connected to the tank **27** via the bypass cut-off valve **49R**. Furthermore, the turning directional control valve **32** and a directional control valve (for example, an arm directional control valve and a second boom directional control valve) not illustrated in the drawings are provided in the right bypass passage **40R**, on the upstream side of the bypass cut-off valve **49R**, but on the downstream side of the right traveling directional control valve **31R**, and the degree of opening of the right bypass passage **40R** is adjusted according to the position of the spool of each directional control valve including the directional control valve **32**.

(127) In the hydraulic drive system configured as described above, the hydraulic-pump flow-rate calibration process is performed using the bypass cut-off valves **49L**, **49R** as the exhaust valves. Specifically, in Step **S1**, the bypass cut-off valve **49L** is opened to connect the left supply passage **34L** to the tank **27** via the left bypass passage **40L**, and the entire amount of the operating fluid dispensed from the left hydraulic pump **21L** returns to the tank **27**. On the other hand, the right bypass passage **40R** is closed by the spool **32a** of the turning directional control valve **32** regardless of whether the bypass cut-off valve **49R** is open or closed. Furthermore, in Step **S7**, the bypass cut-off valve **49L** is closed to keep the operating fluid from flowing back from the left supply passage **34L** to the tank **27**. Thus, by using the bypass cut-off valve **49L** located in the bypass passage **40L**, it is possible to achieve the hydraulic-pump flow-rate calibration process without the unloader valves **45L**, **45R**. Note that even in the case where the unloader valves **45L**, **45R** are provided, it is possible to perform the hydraulic-pump flow-rate calibration process by substantially the same method without operating the unloader valves **45L**, **45R**.

(128) Furthermore, in the hydraulic drive system **1A** according to Embodiment 2, the minimum flow rate Q_{min} is used as the correction flow rate, but this does not necessarily need to be the case; it is sufficient that the flow rate to be used be a known flow rate. Moreover, the outflow rate does not necessarily need to be calculated using Expression 1 mentioned above; the flow rate sensor may be connected to the replenishing passage **47a** to directly detect the outflow rate.

(129) From the foregoing description, many modifications and other embodiments of the present invention would be obvious to a person having ordinary skill in the art. Therefore, the foregoing description should be interpreted only as an example and is provided for the purpose of teaching the best mode for carrying out the present invention to a person having ordinary skill in the art. Substantial changes in details of the structures and/or functions of the present invention are possible within the spirit of the present invention.

REFERENCE CHARACTERS LIST

(130) **1**, **1A-1E** hydraulic drive system (hydraulic-pump flow-rate calibration system) **11L** left traveling hydraulic motor **12** turning hydraulic motor **13** boom cylinder **14** arm cylinder **15** bucket cylinder **21L** left hydraulic pump **21R** right hydraulic pump **23D**, **23L**, **23R** regulator **27** tank **30** straight travel valve (switch valve) **32** turning directional control valve (exhaust valve) **33R** right pump passage **34R** right supply passage **40R** right bypass passage **44** check valve (bypass check valve) **45R** right unloader valve (exhaust valve) **47** replenishing unit **47b** throttle **50**, **50A**, **50B**, **50C**, **50D** control unit (control device, calibration device) **60** gyroscope sensor **60D** rotation sensor **62D** pressure sensor **62R** right pressure sensor **62L** left pressure sensor

Claims

1. A hydraulic-pump flow-rate calibration system, comprising: a hydraulic pump of a variable capacitance type that is connected to a hydraulic actuator and supplies an operating fluid to the hydraulic actuator, the hydraulic actuator operating at a speed corresponding to a flow rate of the

operating fluid supplied to the hydraulic actuator; a regulator that changes a dispense flow rate of the hydraulic pump according to a flow rate command signal input to the regulator; a flow rate detection device that detects a flow rate of the operating fluid to be supplied to the hydraulic actuator; a control device that outputs, to the regulator, the flow rate command signal calculated based on a reference characteristic, to control the regulator; and a calibration device that calculates an actual measurement characteristic indicating the dispense flow rate for the flow rate command signal, and performs, on the reference characteristic, calibration based on the actual measurement characteristic, wherein the actual measurement characteristic is calculated based on the flow rate detected, as a result of the flow rate of the operating fluid to be supplied to the hydraulic actuator being detected by the flow rate detection device during output of a predetermined flow rate command signal from the control device to the regulator, and the control device outputs, to the regulator, the flow rate command signal calculated based on the reference characteristic calibrated based on the actual measurement characteristic, to control the dispense flow rate of the hydraulic pump.

2. The hydraulic-pump flow-rate calibration system according to claim 1, wherein: the hydraulic actuator is a hydraulic motor; and the flow rate detection device includes a rotation sensor that detects a value corresponding to a rotational speed of an output shaft of the hydraulic motor and detects, on the basis of a result of the detection of the rotation sensor and a displacement of the hydraulic motor, the flow rate of the operating fluid to be supplied to the hydraulic motor.

3. The hydraulic-pump flow-rate calibration system according to claim 2, wherein: the hydraulic motor causes a turning body rotatably provided on a structure to turn; the rotation sensor detects a speed of turning of the turning body as the value corresponding to the rotational speed of the output shaft of the hydraulic motor; and the flow rate detection device detects, on the basis of the speed of turning detected and the displacement of the hydraulic motor, the flow rate of the operating fluid to be supplied to the hydraulic motor.

4. The hydraulic-pump flow-rate calibration system according to claim 3, further comprising: a control unit that includes the calibration device and is provided on the turning body, wherein: the rotation sensor is a gyroscope sensor and is embedded in the control unit.

5. A hydraulic-pump flow-rate calibration system, comprising: a first hydraulic pump of a variable capacitance type that is connected to a hydraulic actuator and supplies an operating fluid to the hydraulic actuator, the hydraulic actuator operating at a speed corresponding to a flow rate of the operating fluid supplied to the hydraulic actuator; a second hydraulic pump that is connected to the hydraulic actuator and supplies the operating fluid to the hydraulic actuator; a first regulator that changes a dispense flow rate of the first hydraulic pump according to a first flow rate command signal input to the first regulator; a switch valve that is connected to the first hydraulic pump, the second hydraulic pump, and the hydraulic actuator and connects one of the first hydraulic pump and the second hydraulic pump to the hydraulic actuator; a flow rate detection device that detects the flow rate of the operating fluid to be supplied to the hydraulic actuator; a control device that outputs, to the first regulator, the first flow rate command signal calculated based on a first reference characteristic, to control the first regulator; and a calibration device that calculates a first actual measurement characteristic indicating the dispense flow rate of the first hydraulic pump for the first flow rate command signal, and performs, on the first reference characteristic, calibration based on the first actual measurement characteristic, wherein the first actual measurement characteristic is calculated based on the flow rate detected, as a result of the first hydraulic pump and the hydraulic actuator being connected by the switch valve and the flow rate of the operating fluid to be supplied to the hydraulic actuator being detected by the flow rate detection device during output of a predetermined first flow rate command signal from the control device to the first regulator, and the control device outputs, to the first regulator, the first flow rate command signal calculated based on the first reference characteristic calibrated based on the first actual measurement characteristic, to control the dispense flow rate of the first hydraulic pump.

6. The hydraulic-pump flow-rate calibration system according to claim 5, wherein: the hydraulic actuator is a hydraulic motor; and the flow rate detection device includes a rotation sensor that detects a value corresponding to a rotational speed of an output shaft of the hydraulic motor and detects, on the basis of a result of the detection of the rotation sensor and a displacement of the hydraulic motor, the flow rate of the operating fluid to be supplied to the hydraulic motor.

7. The hydraulic-pump flow-rate calibration system according to claim 6, wherein: the hydraulic motor causes a turning body rotatably provided on a structure to turn; the rotation sensor detects a speed of turning of the turning body as the value corresponding to the rotational speed of the output shaft of the hydraulic motor; and the flow rate detection device detects, on the basis of the speed of turning detected and the displacement of the hydraulic motor, the flow rate of the operating fluid to be supplied to the hydraulic motor.

8. The hydraulic-pump flow-rate calibration system according to claim 7, further comprising: a control unit that includes the calibration device and is provided on the turning body, wherein: the rotation sensor is a gyroscope sensor and is embedded in the control unit.

9. The hydraulic-pump flow-rate calibration system according to claim 5, further comprising: a second regulator that changes, according to a second flow rate command signal input to the second regulator, a dispense flow rate of the second hydraulic pump that is of the variable capacitance type, wherein: the control device outputs the second flow rate command signal to the second regulator to control the second regulator; the calibration device calculates a second actual measurement characteristic of the dispense flow rate of the second hydraulic pump for the second flow rate command signal, and performs, on a second reference characteristic, calibration based on the second actual measurement characteristic; and the second actual measurement characteristic is calculated as a result of the second hydraulic pump and the hydraulic actuator being connected by the switch valve and the flow rate of the operating fluid to be supplied to the hydraulic actuator being detected by the flow rate detection device during output of a predetermined second flow rate command signal to the second regulator.

10. The hydraulic-pump flow-rate calibration system according to claim 9, further comprising: a replenishing unit connected to each of a supply passage formed between a first hydraulic actuator and the switch valve and a pump passage formed between the first hydraulic pump and the switch valve, the first hydraulic actuator being the hydraulic actuator; an exhaust valve connected to the pump passage and configured to be openable and closable, the exhaust valve being opened to discharge, to a tank, the operating fluid flowing in the pump passage; and an outflow rate detection device that detects a flow rate of the operating fluid flowing through the replenishing unit, wherein: the switch valve is further connected to a second hydraulic actuator different from the first hydraulic actuator, and when the first hydraulic pump is connected to the first hydraulic actuator, the switch valve connects the second hydraulic pump to the second hydraulic actuator, and when the second hydraulic pump is connected to the first hydraulic actuator, the switch valve connects the first hydraulic pump to the second hydraulic actuator; when the second hydraulic pump is connected to the first hydraulic actuator by the switch valve, the replenishing unit allows a flow directed from the supply passage to the pump passage to replenish the second hydraulic actuator with the operating fluid dispensed from the second hydraulic pump and blocks an opposite flow of the operating fluid; the first actual measurement characteristic is calculated as a result of the first hydraulic pump and the first hydraulic actuator being connected by the switch valve and the flow rate of the operating fluid to be supplied to the first hydraulic actuator when the exhaust valve is closed being detected by the flow rate detection device during the output of the predetermined first flow rate command signal from the control device to the first regulator; and the second actual measurement characteristic is calculated on the basis of the flow rate detected by the flow rate detection device and an outflow rate detected by the outflow rate detection device, as a result of the second hydraulic pump and the first hydraulic actuator being connected by the switch valve and the flow rate of the operating fluid to be supplied to the first hydraulic actuator when the exhaust valve

is open being detected by the flow rate detection device during the output of the predetermined second flow rate command signal to the second regulator.

11. The hydraulic-pump flow-rate calibration system according to claim 10, wherein: the replenishing unit includes a throttle; and the outflow rate detection device includes a first pressure sensor that detects an outlet pressure of the first hydraulic pump and a second pressure sensor that detects an outlet pressure of the second hydraulic pump, and calculates the outflow rate on the basis of a difference between pressures detected by the first pressure sensor and the second pressure sensor.

12. The hydraulic-pump flow-rate calibration system according to claim 5, further comprising: a second regulator that changes, according to a second flow rate command signal input to the second regulator, a dispense flow rate of the second hydraulic pump that is of the variable capacitance type; and a bypass passage connecting a supply passage formed between a first hydraulic actuator and the switch valve and a pump passage formed between the first hydraulic pump and the switch valve, the bypass passage including a bypass check valve that blocks a flow directed from the supply passage to the pump passage, the first hydraulic actuator being the hydraulic actuator, wherein: the switch valve is further connected to a second hydraulic actuator different from the first hydraulic actuator, and when the first hydraulic pump is connected to the first hydraulic actuator, the switch valve connects the second hydraulic pump to the second hydraulic actuator, and when the second hydraulic pump is connected to the first hydraulic actuator, the switch valve connects the first hydraulic pump to the second hydraulic actuator; the control device outputs the second flow rate command signal to the second regulator to control the second regulator; the calibration device calculates a second actual measurement characteristic of the dispense flow rate of the second hydraulic pump for the second flow rate command signal, and performs, on a second reference characteristic, calibration based on the second actual measurement characteristic; the second actual measurement characteristic is calculated on the basis of a detection flow rate and a correction flow rate detected by the flow rate detection device, as a result of the first flow rate command signal serving as a reference being output to the first regulator, the second hydraulic pump being connected to the first hydraulic actuator by the switch valve, the operating fluid dispensed from the first hydraulic pump being supplied to the first hydraulic actuator via the bypass passage, the operating fluid dispensed from the second hydraulic pump being supplied to the first hydraulic actuator via the switch valve, and the flow rate of the operating fluid to be supplied to the first hydraulic actuator being detected by the flow rate detection device during output of a predetermined second flow rate command signal to the second regulator; and the correction flow rate is detected by the flow rate detection device when the first flow rate command signal serving as the reference is output from the control device to the first regulator and the first hydraulic pump is connected to the first hydraulic actuator by the switch valve.

13. The hydraulic-pump flow-rate calibration system according to claim 9, wherein: the switch valve is capable of connecting both the first hydraulic pump and the second hydraulic pump to the hydraulic actuator; the calibration device calculates the second actual measurement characteristic of the dispense flow rate of the second hydraulic pump for the second flow rate command signal, and performs, on a second reference characteristic, calibration based on the second actual measurement characteristic; the second actual measurement characteristic is calculated on the basis of a detection flow rate and a correction flow rate detected by the flow rate detection device, as a result of the first flow rate command signal serving as a reference being output to the first regulator, both the first hydraulic pump and the second hydraulic pump being connected to the hydraulic actuator by the switch valve, and the flow rate of the operating fluid to be supplied to the hydraulic actuator being detected by the flow rate detection device during the output of the predetermined second flow rate command signal to the second regulator; and the correction flow rate is a flow rate of the operating fluid flowing through the hydraulic actuator when the first flow rate command signal serving as the reference is output from the control device to the first regulator

and the first hydraulic pump is connected to the hydraulic actuator by the switch valve.

14. The hydraulic-pump flow-rate calibration system according to claim 1, wherein: the calibration device corrects, on the basis of an amount of leakage at the hydraulic actuator, the flow rate detected by the flow rate detection device, and calculates the actual measurement characteristic on the basis of the flow rate corrected.

15. The hydraulic-pump flow-rate calibration system according to claim 1, wherein: the actual measurement characteristic is calculated on the basis of a plurality of flow rates detected by the flow rate detection device when a plurality of flow rate command signals different from each other are output.

16. The hydraulic-pump flow-rate calibration system according to claim 1, wherein: when a predetermined condition is met, the calibration device calculates the actual measurement characteristic.
